Lecture Notes on QFT I

December 14, 2025

Readings

- Peskin & Schroeder Sec. 2.1 and 2.2
- Weinberg vol 1 Chap. 1

Lecture 1: Principle of Locality and Fields

The goals of this Quantum Field Theory class is

- To develop concepts, formalism and techniques for quantum dynamics of fields.
- Quantum Field Theory is also very important for other interactions in nature. Among the four fundamental interactions in nature, three of them are described by QFT completely.
- Even though QFT is initially developed for particle physics, but over the years, QFT has been found many applications in many other branches of physics, in condensed matter, statistical physics and so on. It's fair to say, nowadays, QFT has become a universal language for theoretical physics in all different fields.

Cahpter 1. Why we consider QFT?

principle of locality

The first important concept is called the **principle of locality**.

In Newtonian mechanics, you have action at a distance, for example, if you look at gravity and the gravity is exerted by the sun on the Earth, but they are very far away. And the same thing with the Coulomb interaction between the charged particles.

But then, in the 19th century, they came from this principle of locality, that is formulated by Farady around 1830. The principle of locality says that all points you don't have action at a distance. He said **all points in space participate in the physical process, and the effect propagates from points to a neighboring points**. So in this principle of locality, you don't have action at a distance, action at a distance is always conveyed from one point to another point through the propagating in the space.

The concepts of **fields** is **the mathematical device or vehicle that the principle locality is at work**. The main idea of the filed is that we associate each point in space a dynamical variable.

For example, if you have electric field is defined at each point x, we can introduce an electric field, which can also depend on time. Of course, normally we write it this way $\vec{E}_{\vec{x}}(t) \equiv \vec{E}(\vec{x}, t)$. The reason I write it in this way $\vec{E}_{\vec{x}}(t)$ to emphasize that in the definition of the electric field, the space and times actually play very different roles. The space plays the role of a label. So at each point $\vec{x}$, we have electric field, so x here is just a label, and t is used to describe the evolution, the change of the electric field. So $\vec{x}$ and t play very different roles.

Similarly, you can do this to magnetic field $\vec{B}_{\vec{x}}(t) \equiv \vec{B}(\vec{x}, t)$, you should always treat $\vec{x}$ as labels. And we know the evolution of electric and magnetic field is described by Maxwell's equations,

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi \vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases} \tag{1}$$

which is called the differential form of the Maxwell's equations. This set of equations examplifies perfectly the principle of locality. It's because you see those equations only involve the value of electric fields and the magnetic field at a single point. If $\vec{B}$ is at $\vec{x}$ point, then $\vec{E}$ is also at $\vec{x}$ point. You never say $\vec{B}_{\vec{x}}$ but $\vec{E}_{\vec{y}}$. So this reflects the principle of locality that the effect is propagating from point to point, you just have the derivative of the same point. You don't involve separate points. So we call Maxwell's equations as local equations, they contain only E and B at same point with finite number of derivatives. So the derivatives are the ones to help you to propagate because it relates the point to the neighboring point.

Another example of locality is the Einstein's general relativity. In Einstein gravity, the dynamical variables are so-called spacetime metric $g_{\mu\nu}(\vec{x}, t)$. They are object with two indices, two spacetime indices, and the Einstein's equations are equations for this kind of object $g_{\mu\nu}$. And again this equation are local equations, in the sense that they only depends on the g evaluated at the same point with a finite number of derivatives. So in Einstein's gravity, you no longer have action at distance, so the effect of the gravity is propagated through the spacetime.

Here we say these two examples examplify the principle of locality. In fact, the principle of locality plays a very important role in formulating those equations, which is a very powerful principle.

Lecture 2: Classical Field Theories

Types of field

- A scalar field $\phi(\vec{x}, t)$. This is a quantity defined at each point; there is only one value at the given point. For example, the temperature is a single scalar quantity defined at each point.
- A vector field: $\vec{E}(t, \vec{x})$ and $\vec{B}(t, \vec{x})$ are examples of vector fields because you have a vector at each point. Here, I'm using the three-dimensional notations, but in the relativistic theory, we can use the four vector notations, e.x., the vector potential has the form $A_\mu(x^\nu)$. Here A_μ is a four-vector, obtained by combining the spatial and temporal components into a single spacetime vector.
- A tensor field: $g_{\mu\nu}(x^\lambda)$ is an example for tensor field. Here you have two indices, you have many different components depend on the dimension. At each point, you have an object with two indices.
- Spinor fields $\psi_\alpha(t, \vec{x})$, where α is another index that we will define later.

Convention

- μ is always from 0 to 3, and sometimes I write simply x; when I write $x = x^\mu = (ct, \vec{x}) = (ct, x^i)$, it denotes a four-vector. The spatial index is always denoted by i.
- $\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$, ∇ is the gradient on the spatial directions.
- A four-vector $A_\mu = (A_0, \vec{A}) = (A_0, A_i)$, we still use i to denote the spatial components.
- $c, \hbar = 1$.

Action Principle in Classical Mechanics

First, recall in your classical mechanics, we introduce the action

$$S = \int \mathrm{d}t L(x, \dot{x}, t), \tag{2}$$

where Lagrangian is a function of x, $\dot{x}$ and the time. This is the one-dimensional particle motion. And you can also introduce the canonical momentum, which is defined by

$$p = \frac{\partial L}{\partial \dot{x}}. \tag{3}$$

The Hamiltonian is related to the Lagrangian by

$$H = p\dot{x} - L, \tag{4}$$

through a Legendre transformation. And the equation of motion is obtained by extremizing $S[x(t)]$, where S can be consider as a functional of your trajectory, so whatever trajectory you have, you extremize this S, and you obtain the equation of motion (EOM).

Classical Field Theory

Now we can generalize this to field theory. From the principle of locality, the form of the Lagrangian L in field theory is significantly constrained. It must have the form:

$$L = \int \mathrm{d}^3x \mathcal{L}\left(\phi_a, \partial_t\phi_a, \partial_i\phi_a\right) \tag{5}$$

which is the integration over all spatial directions. Here, $\phi_a(\vec{x}, t)$ is a function of spatial direction and time, it is a general field, and a label different fields. e.g., a can label different scalar fields if you have different scalar fields, a can also refer to A_μ or to spinor fields, and so on. $\mathcal{L}$ is called the Lagrangian density, which depends only on the value of ϕ_a and its derivatives at a single point. They are still local.

Remarks: Why the Lagrangian must have this form?

- First, the principle of locality implies (228) only involves a single integral because the locality does not allow something like this,
$$L = \cdots + \int \mathrm{d}^3x\mathrm{d}^3y\phi(\vec{x},t)\phi(\vec{y},t) \tag{6}$$
which involving the ϕ at different points. If your Lagrangian contains such terms, then when we derive the equations of motion, they will not be local. EOM will involve the behavior of your field at one point and then influenced by the field value at a distant point, then you will not have local equation.

- We only allow first derivative in time, we don't allow high order derivatives in time. The reason is that again the equation of motion will involve more than two derivatives in time if your Lagrangian has high order derivatives in time. $\partial_t\phi$ in (228) implies that EOMs contain two t-derivatives, which come from our experiences. In real life, all experiments are determined by the initial conditions. For the initial condition, you only need to specify the position and velocity; no higher derivatives are required. If the equation of motion involves more than two derivatives in time, then you need to specify more general initial conditions. But for simplicity, for the most of the time, as we will see, we will restrict to quantum filed theory in first derivative in time.

So now as in classical mechanics, we can introduce the canonical momentum, Hamiltonian, and so on. Here, we can introduce the so-called canonical momentum density
$$\Pi_a(\vec{x},t) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}_a(\vec{x},t)}, \quad \text{where} \quad \dot{\phi}_a(\vec{x},t) = \partial_t\phi_a \tag{7}$$
So this is just the direct generalization of classical mechanics. The momentum density is evaluated at each point, so Π_a also depends on $\vec{x}$ and t. The reason we call this the momentum density is because here $\mathcal{L}$ is the Lagrangian density. In the same way, we can also define the Hamiltonian density associated with each degree of freedom (DOF):
$$\mathcal{H} = \Pi_a\dot{\phi}_a - \mathcal{L} \tag{8}$$
Then, the Hamiltonian you just integrate over all space
$$H = \int \mathrm{d}^3x\mathcal{H} \tag{9}$$
essentially a sum over all degrees of freedom. So now we can talk about equation of motion. It is analogous to classical mechanics. The classical field theory is just like classical mechanics, you generalize it to infinite number of DOFs. So now with each point in space, you associate it with some DOFs. Again, you just do the variation of the action, you extremize the action to be 0,
$$\delta S = 0 \tag{10}$$
where the action is again defined by
$$S = \int \mathrm{d}tL = \int \mathrm{d}^4x\mathcal{L}(\phi_a, \partial_\mu\phi_a). \tag{11}$$
You can straightforwardly generalize this to include higher spatial derivatives. Now let us compute the variation:
$$0 = \delta S = \int \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial\left(\partial_\mu\phi_a\right)}\delta\left(\partial_\mu\phi_a\right)\right] = \int \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial\left(\partial_\mu\phi_a\right)}\partial_\mu\left(\delta\phi_a\right)\right] \tag{12}$$
In this case, δ is the variation, and ∂_μ and δ are independent, so you can exchange them, i.e.,
$$\partial_\mu\left(\delta\phi_a\right) = \delta\left(\partial_\mu\phi_a\right) \tag{13}$$
Then, we can do the integration by part for the second term and obtain
$$0 = \int \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\left(\partial_\mu\phi_a\right)}\right)\delta\phi_a\right] + \text{boundary terms} \tag{14}$$
The boundary terms will be proportional to $\delta\phi_a$. We will always choose $\delta\phi_a$ such that boundary terms vanish. Equation (14) must vanish for arbitrary variations $\delta\phi_a(x^\mu)$, which can be any function of x^μ. Therefore, the prefactor has to vanish, i.e.,
$$0 = \frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\left(\partial_\mu\phi_a\right)}\right), \tag{15}$$
(15) is the general EOM for classical field theory.

Restrictions in Field Theory

From now on, we will make two restrictions for simplicity.

Translationally invariant

The first is that we restrict to field theory which are translationally invariant, which means the theory does not single out any special point in spacetime. If you perform the experiment here in Chengdu, it is the same as performing it in Washington DC. If the theory were not translationally invariant, experiments performed in Chengdu and Washington DC would have different results.

Lorentz invariant

The second is that we assume the field theory is Lorentz invariant. A Lorentz invariant quantity looks exactly the same for every observer moving at constant velocity, no matter how their coordinate axes are rotated or how fast they move relative to each other. A quantity is Lorentz invariant if it does not change under any Lorentz transformation that includes both: Spatial rotations (changing direction in space), and Lorentz boosts (changing velocity or reference frame).

Now let us look at some simple examples of classical field theories that satisfy these two conditions.

E.x. 1: Maxwell's theory.

The dynamical variable is the four-vector potential $A_\mu(x^\nu)$. For A_μ, you define the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{16}$$

The electric and magnetic fields E and B can be obtained from $F_{\mu\nu}$. The action for E&M can be obtained by

$$S_{EM} = \int \mathrm{d}^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \tag{17}$$

This is the action for electromagnetism in the absence of charged matter. If you do the variation here to get the EOM, then you obtain the source-free (vacuum) Maxwell equations. This action is corresponding to a local theory due to a single integral of spacetime.

Lecture 3: Symmetry and Conservation Law

Ex 2: Einstein gravity Einstein gravity is another classical theory. The Einstein-Hilbert action is

$$S = \frac{1}{16\pi G_N}\int \mathrm{d}^4x\sqrt{-g}R, \tag{18}$$

where g is the determinant of the metric and R is the Ricci scalar. Einstein gravity is a local theory involving a single spacetime integral over geometric quantities.

Ex 3: The simplest quantum field theory the scalar field theory: A_μ is the vector field in E&M, and in Einstein gravity the dynamical variable is the metric tensor $g_{\mu\nu}$, which is even more complicated. Thus, the simplest case to study is a scalar field.

Let us just consider the real-valued scalar field $\phi(x)$. At each spacetime point, the field takes a single value; thus $\phi(x)$ is the dynamical variable. In physics, there are many examples of scalar fields; e.g., the Higgs field or the pion field, whose excitations correspond to scalar particles.

The simplest family of scalar field actions has the following form:

$$S = \int \mathrm{d}^4x\left[-\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)\right], \tag{19}$$

which is compatible with locality, translational symmetry, and Lorentz symmetry. Here, you can see that the Lorentz indices are contracted. The Lorentz indices in the derivative are contracted, so this guarantees the theory Lorentz invariant. This is essentially the simplest quantum theory one can write down based on locality.

In general, we can take $V(\phi)$ to be a polynomial

$$V(\phi) = \text{function of } \phi \tag{20}$$

When a system has translational symmetry, it means that if you perform an experiment anywhere, the result will be the same. This implies that all parameters in $V(\phi)$ must be constant and cannot depend on spacetime. The canonical momentum is

$$\Pi(x) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi} \tag{21}$$

where $\dot{\phi}^2 = \partial_t^2\phi$ due to

$$\partial_\mu\phi\partial^\mu\phi = -\dot{\phi}^2 + (\nabla\phi)^2 \tag{22}$$

And then the Hamiltonian density is

$$\mathcal{H} = \Pi\dot{\phi} - \mathcal{L} = \frac{\Pi^2}{2} + \frac{1}{2}(\nabla\phi)^2 + V(\phi), \tag{23}$$

which is expressed in terms of momentum. And EOM is given by

$$\partial^2\phi - \frac{\partial V}{\partial\phi} = 0. \tag{24}$$

and again I'm using a shorthand notation,

$$\partial^2 = \partial_\mu\partial^\mu = -\partial_t^2 + \nabla^2. \tag{25}$$

Here, let me give some simple examples of the V. In the simplest case, take $V = f\phi$. From translational invariance, f must be constant and cannot depend on spacetime. Then, the EOM is

$$\partial^2\phi = f, \tag{26}$$

In this case, essentially you have some kind of constant, which is source for this ϕ. In this case, f acts as a constant source for the field ϕ. We say the system has an external force f; because if $f \neq 0$, then ϕ cannot vanish. In this sense, the field ϕ is always generated by f. However, we usually consider systems without external forces, where the field evolves on its own.

In that case, if we ignore external forces, the simplest situation is

$$V(\phi) = \frac{1}{2}m^2\phi^2 \tag{27}$$

m has to be constant. In this case, you have the EOM:

$$\partial^2\phi - m^2\phi = 0, \tag{28}$$

This is the famous equation called the Klein-Gordon equation in classical field theory. And you can also consider more complicated field theories with $V(\phi)$ to be some higher polynomial or more complicated functions. But keep in mind, this square is special as then the EOM has linear term. However, when $V(\phi)$ includes terms beyond quadratic order, the equation becomes nonlinear, and the analysis becomes more complicated. Anyway, this would be the simplest theory we consider.

Symmetry

A **symmetry** is a transformation of the dynamical fields that leaves the action invariant. Formally, each field ϕ_a transforms into a function of all the fields:

$$\phi_a \longrightarrow \phi'_a(\phi_b), \tag{29}$$

and if the action S remains unchanged under this transformation, we say the theory has a symmetry.

Example: Translation symmetry. Consider the scalar field theory introduced in (19). If we perform a constant shift of spacetime coordinates,

$$x^\mu \longrightarrow x'^\mu = x^\mu + a^\mu, \tag{30}$$

where a^μ is a constant four-vector, the scalar field transforms as

$$\phi(x) = \phi'(x'), \tag{31}$$

meaning that the field value at a given spacetime point does not change, only its coordinate label does. You can verify that the scalar-field action is invariant under this translation, confirming that translation is indeed a symmetry.

Example: Lorentz symmetry. Similarly, a Lorentz transformation acts on spacetime as

$$x^\mu \longrightarrow x'^\mu = \Lambda^\mu{}_\nu\, x^\nu, \tag{32}$$

where $\Lambda^\mu{}_\nu$ is a constant Lorentz transformation matrix satisfying $\Lambda^{\mathrm{T}}\eta\,\Lambda = \eta$. Under this transformation, the scalar field again obeys (31). The action remains invariant, so Lorentz invariance is another continuous spacetime symmetry.

Continuous and discrete symmetries. Transformations such as translations and Lorentz transformations depend on continuous parameters (a^μ and $\Lambda^\mu{}_\nu$), so they are called **continuous symmetries**. In four dimensions, a^μ has four continuous parameters, while $\Lambda^\mu{}_\nu$ contains six (three rotations and three boosts).

The same scalar-field action also admits **discrete symmetries**, for example:

$$\phi \longrightarrow -\phi, \tag{33}$$
$$\vec{x} \longrightarrow -\vec{x}, \tag{34}$$

corresponding to field inversion and spatial parity. Both leave the action invariant but involve no continuous parameters. Because performing either transformation twice returns the system to its original state, each corresponds to a $\mathbb{Z}_2$ symmetry.

Global and local symmetries. Symmetries are further classified as **global** or **local**:

- *Global symmetries* have spacetime-independent transformation parameters. Translations and Lorentz transformations are examples, since a^μ and $\Lambda^\mu{}_\nu$ are constants.
- *Local symmetries* have spacetime-dependent parameters. A well-known example is the gauge symmetry in electromagnetism, where the phase transformation depends on spacetime position.

Noethers theorem. A deep connection exists between continuous symmetries and conservation laws: each continuous global symmetry corresponds to a conserved quantity. In classical mechanics, time translation implies energy conservation, spatial translation implies momentum conservation, and rotational symmetry implies angular momentum conservation. This general relationship is formalized by **Noethers theorem**.

Consider a continuous symmetry with an infinitesimal parameter ϵ:

$$\phi_a \longrightarrow \phi'_a = \phi_a + \epsilon\, f_a(\phi_b, \partial_\mu\phi_b), \tag{35}$$

where f_a is a function of the fields and possibly their derivatives. The variation of the Lagrangian density under this infinitesimal transformation must be a total derivative:

$$\hat{\delta}\mathcal{L} = \epsilon\,\partial_\mu K^\mu, \tag{36}$$

for some current K^μ. If $K^\mu = 0$, the Lagrangian is strictly invariant.

Using the Euler-Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\right) = 0,$$

the variation of $\mathcal{L}$ can be written as

$$\hat{\delta}\mathcal{L} = \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,\hat{\delta}\phi_a\right) = \epsilon\,\partial_\mu K^\mu. \tag{37}$$

Comparing both sides yields the conservation equation

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,f_a - K^\mu\right) = 0. \tag{38}$$

Defining the **Noether current**

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,f_a - K^\mu, \tag{39}$$

we obtain the conservation law

$$\partial_\mu J^\mu = 0. \tag{40}$$

Continuity equation and conserved charge. Writing $J^\mu = (J^0, \vec{J})$, the conservation law takes the familiar form

$$\partial_0 J^0 + \nabla\cdot\vec{J} = 0, \tag{41}$$

which expresses the local conservation of a density and its associated current. Integrating over all space gives

$$Q = \int \mathrm{d}^3x\, J^0, \tag{42}$$

and differentiating (41) shows

$$\partial_0 Q = 0, \tag{43}$$

so the total charge Q is conserved. Equation (43) is the integrated form of the local continuity equation (41).

Remarks. This result is known as **Noethers first theorem**, valid when ϵ is spacetime independent (global symmetry). When the transformation parameter depends on spacetime, as in gauge theories, additional terms appear and the statement becomes **Noethers second theorem**, which relates local symmetries to redundancies among the equations of motion.

The key idea is that any finite continuous symmetry can be constructed by successively applying infinitesimal transformations. Hence, understanding the infinitesimal form of a symmetry is sufficient to determine its full finite structure.

Lecture 4 Why QFT?

With these preparations, we are ready to begin our discussion of **Quantum Mechanics**.

From Classical Mechanics to Quantum Mechanics

Recall that in **classical mechanics**, the simplest system consists of a single degree of freedom — a particle moving along one dimension, whose position is described by a function $x(t)$. The goal of classical mechanics is to *solve the equation of motion (EOM)* for $x(t)$. Once we find $x(t)$, we have completely described the dynamics of the system.

Quantization: From Variables to Operators

In **quantum mechanics**, the dynamical variable $x(t)$ becomes an **operator**

$$x(t) \longrightarrow \hat{x}(t),$$

and similarly, the conjugate momentum becomes $\hat{p}(t)$. The classical EOMs are replaced by **Heisenberg equations of motion** for these operators.

There are two equivalent but conceptually different pictures of quantum mechanics:

(a) Schrdinger Picture

- The **state** of the system is described by a wave function $\psi(x, t)$.
- Operators such as $\hat{x}$ and $\hat{p}$ are *time-independent.*
- The quantity x in $\psi(x, t)$ is the eigenvalue of $\hat{x}$.
- The time evolution of the system is governed by the **Schrdinger equation**.
- Once we know $\psi(x, t)$, we can compute expectation values and transition amplitudes.

(b) Heisenberg Picture

- The **operators** $\hat{x}(t)$ and $\hat{p}(t)$ are time-dependent.
- The **state vector** $|\psi\rangle$ is time-independent.
- Time evolution is governed by the **Heisenberg equation**

$$\frac{\mathrm{d}\hat{O}}{\mathrm{d}t} = i[\hat{H}, \hat{O}],$$

  where $\hat{H}$ is the Hamiltonian.
- Once we solve the operator equations, expectation values can be evaluated in any state.

From Quantum Mechanics to Quantum Field Theory

In **field theory**, we proceed in an analogous way. The classical dynamical variables are now fields $\phi(\vec{x}, t)$, and upon quantization they become operators $\hat{\phi}(\vec{x}, t)$.

The key conceptual difference is that in classical mechanics, x is a *dynamical variable*, whereas in field theory, $\vec{x}$ merely labels spatial points — the *field* ϕ is the true dynamical variable.

In the Schrdinger picture of field theory:

$$\Psi[\phi(\vec{x}), t]$$

is a *wave functional* — a function of all possible field configurations $\phi(\vec{x})$. Here, $\phi(\vec{x})$ denotes the eigenvalues of the operator $\hat{\phi}(\vec{x})$, which does not depend on time in this picture. The Schrdinger equation determines the time evolution of $\Psi[\phi(\vec{x}), t]$.

In the Heisenberg picture, we focus instead on the operator equations of motion:

$$\hat{\phi}(\vec{x}, t), \quad \hat{\pi}(\vec{x}, t),$$

which satisfy the Heisenberg equations — the quantum analogues of the classical field equations. The state $|\psi\rangle$ remains fixed, and all dynamics are carried by the operators.

Remarks: In Quantum Field Theory, the Heisenberg picture is almost always used, because it directly generalizes classical field equations and simplifies computations. We will therefore adopt the Heisenberg picture from now on.

Why Relativistic Quantum Mechanics Fails

Before introducing QFT, let us comment briefly on **relativistic quantum mechanics**. Naively, one might attempt to combine quantum mechanics and special relativity directly. However, this approach fails for fundamental reasons, leading us naturally to quantum field theory.

From Schrdinger to Klein–Gordon

In non-relativistic quantum mechanics, we start from the dispersion relation

$$E = \frac{\vec{p}^2}{2m}. \tag{44}$$

Promoting energy and momentum to operators,

$$E \to i\frac{\partial}{\partial t}, \qquad \vec{p} \to -i\nabla,$$

we obtain the Schrdinger equation

$$i\frac{\partial}{\partial t}\psi(\vec{x},t) = -\frac{1}{2m}\nabla^2\psi(\vec{x},t). \tag{45}$$

To include relativity, one might instead start with the relativistic dispersion relation

$$E^2 = \vec{p}^2 + m^2. \tag{46}$$

Applying the same operator substitutions yields

$$\partial_\mu\partial^\mu\psi(\vec{x},t) - m^2\psi(\vec{x},t) = 0, \tag{47}$$

the **Klein–Gordon equation**. Historically, this was regarded as the wave equation for a single relativistic particle.

Problems with the Klein–Gordon Interpretation

Soon, physicists realized several problems with interpreting (47) as a one-particle wave equation:

1. **No positive-definite probability density.** The conserved current derived from (47) cannot be interpreted as a non-negative probability density. Probability must be both non-negative and conserved — here, it fails the first condition.

2. **Negative-energy solutions.** From the dispersion relation

   $$E = \pm\sqrt{\vec{p}^2 + m^2},$$

   we see that quantum mechanics cannot simply discard the negative branch. If such states exist, the system becomes unstable: particles would cascade to states of ever-lower (negative) energy.

Deeper Conceptual Problem

Even if these issues were fixed, the entire concept of a wave function for a *fixed number of particles* breaks down in relativity. In non-relativistic systems, particle number is conserved, and a two-particle wave function $\psi(\vec{x}_1, \vec{x}_2, t)$ makes sense. But in relativistic systems, energy can create or annihilate particles: electrons and positrons can annihilate into photons, for example. Thus, the particle number is not fixed, and no single wave function can describe such a system.

The Necessity of Quantum Field Theory

To describe systems where particles can be created and destroyed, we must have a framework that can handle an arbitrary and variable number of particles in a unified way. This is precisely what **quantum field theory** provides.

In QFT, the fundamental objects are fields. Their excitations correspond to particles, automatically allowing for processes of creation and annihilation. Hence, QFT is not merely a generalization of quantum mechanics; it is the only consistent framework that unifies *quantum mechanics* with *special relativity.*

Field Theory from the Continuum Limit of Discrete Systems

The third motivation for Quantum Field Theory comes from the observation that **field theories can also emerge as the continuum limit of discrete systems**. This perspective is especially relevant in **condensed matter physics**.

Example: A Chain of Coupled Oscillators

Consider a one-dimensional chain of particles (for instance, atoms in a crystal lattice) connected by springs. Let the equilibrium spacing between neighboring atoms be a, and label the positions of the atoms by $x_0, x_1, x_2, \ldots$. The displacement of the n-th atom from its equilibrium position is denoted by η_n:

$$x_n - x_n^{(0)} = \eta_n. \tag{48}$$

The Lagrangian of this discrete system is

$$L = T - V = \sum_n \left(\frac{1}{2}\mu\dot{\eta}_n^2 - \frac{\lambda}{2}(\eta_{n+1} - \eta_n)^2 - \frac{1}{2}\sigma\eta_n^2 \right), \tag{49}$$

where

- T is the kinetic energy and V is the potential,
- μ is the mass of each particle,
- λ characterizes the spring coupling between nearest neighbors,
- σ describes the restoring potential trapping each atom near equilibrium.

This is a simple yet instructive model of coupled harmonic oscillators.

Continuum Limit

As the lattice spacing $a \to 0$, the chain becomes a continuous medium. We then replace the discrete displacements $\eta_n(t)$ by a smooth function $\eta(x, t)$, where $x = na$ labels the position along the chain.

In this limit, the discrete sum becomes an integral:

$$a \sum_n \longrightarrow \int \mathrm{d}x. \tag{50}$$

The Lagrangian (49) becomes

$$\begin{aligned} L &= \sum_n a \left(\frac{\mu}{2a}\dot{\eta}_n^2 - \frac{a\lambda}{2}\left(\frac{\eta_{n+1} - \eta_n}{a}\right)^2 - \frac{\sigma}{2a}\eta_n^2 \right) \\ &= \int \mathrm{d}x \left(\frac{\tilde{\mu}}{2}\dot{\eta}^2 - \frac{\tilde{\lambda}}{2}(\partial_x\eta)^2 - \frac{\tilde{\sigma}}{2}\eta^2 \right), \end{aligned} \tag{51}$$

where we define

$$\tilde{\mu} = \frac{\mu}{a}, \qquad \tilde{\lambda} = \frac{\lambda}{a}, \qquad \tilde{\sigma} = \frac{\sigma}{a}. \tag{52}$$

To obtain a well-defined continuum theory, we keep $\tilde{\mu}$, $\tilde{\lambda}$, and $\tilde{\sigma}$ fixed as $a \to 0$.

The Emergent Field Theory

The continuum Lagrangian can be rewritten as

$$L = \tilde{\mu} \int \mathrm{d}x \left(\frac{1}{2}\dot{\eta}^2 - \frac{\tilde{\lambda}}{2\tilde{\mu}}(\partial_x\eta)^2 - \frac{\tilde{\sigma}}{2\tilde{\mu}}\eta^2 \right) = \tilde{\mu} \int \mathrm{d}x \left(\frac{1}{2}\dot{\eta}^2 - \frac{v^2}{2}(\partial_x\eta)^2 - \frac{m^2}{2}\eta^2 \right), \tag{53}$$

where we define

$$v^2 = \frac{\tilde{\lambda}}{\tilde{\mu}}, \qquad m^2 = \frac{\tilde{\sigma}}{\tilde{\mu}}. \tag{54}$$

This Lagrangian is precisely the form of a **scalar field theory** discussed earlier. If we set $v = 1$, it becomes relativistic and corresponds to the case where the propagation speed equals the speed of light. In general, for $v \neq 1$, it describes a **non-relativistic field theory**.

Physical Interpretation

Even though this model is extremely simple, it captures a fundamental idea: many condensed matter systems — such as vibrations in a solid — can be described by lattice models at the microscopic level, but by continuum field theories at macroscopic scales.

When we further include quantum mechanics, the quantization of such continuum fields naturally leads to **quantum field theory**. Thus, QFT is not only a framework for particle physics but also the natural language for describing collective excitations in many-body systems.

Summary: All Paths Lead to QFT

- Quantum Field Theory describes the **quantum dynamics of classical fields** (e.g., electromagnetic, gravitational, scalar fields).
- It **unifies special relativity and quantum mechanics** in a consistent way.
- It also **emerges as the continuum limit of discrete systems**, relevant to many areas such as condensed matter physics.

In short, whether we begin from quantum mechanics, relativity, or discrete many-body systems, **all roads lead to Quantum Field Theory**.

Lecture 5. Quantization of the Harmonic Oscillator and Free Scalar Field

In this lecture, we will discuss how to quantize a theory, starting from the familiar example of the harmonic oscillator and then generalizing to field theory. The key idea is that a free scalar field is essentially an infinite collection of harmonic oscillators labeled by the wave vector $\vec{k}$.

Quantization of the Harmonic Oscillator in the Heisenberg Picture

The Lagrangian for a one-dimensional harmonic oscillator (with mass $m = 1$) is

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \tag{55}$$

The conjugate momentum is $p = \dot{x}$, and the Hamiltonian becomes

$$H = p\dot{x} - L = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \tag{56}$$

The equation of motion is

$$\ddot{x} + \omega^2 x = 0. \tag{57}$$

The general classical solution is

$$x(t) = A\cos(\omega t) + B\sin(\omega t), \tag{58}$$

where A and B are integration constants. Equivalently, we can write it in complex form:

$$x(t) = \frac{1}{\sqrt{2}}\left(ae^{-i\omega t} + a^* e^{i\omega t}\right), \tag{59}$$

where a is a complex integration constant.

Quantization: To quantize the system, we promote the classical variables to quantum operators in the Heisenberg picture:

$$x(t) \to \hat{x}(t), \qquad p(t) \to \hat{p}(t). \tag{60}$$

The operator equation of motion is identical in form to the classical one,

$$\ddot{\hat{x}} + \omega^2 \hat{x} = 0, \tag{61}$$

but its interpretation changes since $\hat{x}$ is now an operator. The general operator solution mirrors the classical one:

$$\hat{x}(t) = \hat{A}\cos(\omega t) + \hat{B}\sin(\omega t) = \frac{1}{\sqrt{2}}\left(\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}\right), \tag{62}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are constant operators (time-independent in the Heisenberg picture). The momentum operator follows from differentiation:

$$\hat{p}(t) = \dot{\hat{x}}(t) = \frac{-i\omega}{\sqrt{2}}\left(\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}\right). \tag{63}$$

Canonical Quantization: We impose the canonical commutation relation

$$[\hat{x}(t), \hat{x}'(t)] = [\hat{p}(t), \hat{p}'(t)] = 0, \qquad [\hat{x}(t), \hat{p}(t)] = i, \tag{64}$$

where we assume $\hbar = 1$. This implies the fundamental algebra for the creation and annihilation operators:

$$[\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}^\dagger] = 0, \qquad [\hat{a}, \hat{a}^\dagger] = 1 \tag{65}$$

Hilbert Space Construction: The vacuum state is defined as

$$\hat{a}|0\rangle = 0, \tag{66}$$

and the excited states are obtained by applying $\hat{a}^\dagger$ repeatedly:

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle. \tag{67}$$

These states form a complete orthonormal basis for the Hilbert space of the harmonic oscillator.

Summary of the Quantization Procedure: The general steps for quantizing a system are:

1. Start with the classical equation of motion.
2. Find the most general classical solution.
3. Promote classical dynamical variables to quantum operators.
4. Impose canonical commutation relations.
5. Use the resulting constant operators to construct the Hilbert space.

This procedure applies equally well to systems with many degrees of freedom, including field theories.

Quantization of the Free Scalar Field

We now apply the same logic to a relativistic field with infinitely many degrees of freedom. Consider the free scalar field with action

$$S = \int \mathrm{d}^4x \left[\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2\right]. \tag{68}$$

The equation of motion is

$$(\partial^2 + m^2)\phi = 0. \tag{69}$$

The canonical momentum and Hamiltonian density are

$$\Pi = \dot{\phi}, \qquad \mathcal{H} = \frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2. \tag{70}$$

Classical Solutions: Because of translation symmetry, we can expand ϕ in Fourier modes:

$$\phi(x) = e^{-iEt+i\vec{k}\cdot\vec{x}}. \tag{71}$$

Substituting into (69) gives the dispersion relation

$$E^2 = \vec{k}^2 + m^2, \tag{72}$$

Here we define the frequency $\omega_{\vec{k}}$

$$\omega_{\vec{k}} \equiv \sqrt{\vec{k}^2 + m^2}. \tag{73}$$

Then, we define the positive-frequency and negative-frequency solutions:

$$u_{\vec{k}}(x) = e^{-i\omega_{\vec{k}}t+i\vec{k}\cdot\vec{x}}, \tag{74}$$

$$u^*_{\vec{k}}(x) = e^{i\omega_{\vec{k}}t-i\vec{k}\cdot\vec{x}}. \tag{75}$$

The most general classical solution is

$$\phi(x) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\left(a_{\vec{k}}u_{\vec{k}}(x) + a^*_{\vec{k}}u^*_{\vec{k}}(x)\right). \tag{76}$$

Quantization: Promote ϕ and Π to operators:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\left(\hat{a}_{\vec{k}}u_{\vec{k}}(x) + \hat{a}^\dagger_{\vec{k}}u^*_{\vec{k}}(x)\right). \tag{77}$$

The canonical commutation relations generalize to

$$[\hat{\phi}(t,\vec{x}), \hat{\phi}(t,\vec{x}')] = 0, \tag{78}$$

$$[\hat{\Pi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')] = 0, \tag{79}$$

$$[\hat{\phi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'). \tag{80}$$

From these, one can derive

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = 0, \tag{81}$$

$$[\hat{a}^\dagger_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = 0, \tag{82}$$

$$[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = (2\pi)^3\delta^{(3)}(\vec{k} - \vec{k}'). \tag{83}$$

Interpretation: Each mode $\vec{k}$ behaves as an independent quantum harmonic oscillator with frequency $\omega_{\vec{k}}$. Therefore, a free scalar field can be viewed as an infinite set of harmonic oscillators, one for each $\vec{k}$.

Hilbert Space: The vacuum state $|0\rangle$ satisfies

$$\hat{a}_{\vec{k}}|0\rangle = 0, \qquad \forall\vec{k}. \tag{84}$$

Excited states are created by acting with $\hat{a}^\dagger_{\vec{k}}$:

$$|\vec{k}_1, \vec{k}_2, \cdots, \vec{k}_n\rangle = \hat{a}^\dagger_{\vec{k}_1}\hat{a}^\dagger_{\vec{k}_2}\cdots\hat{a}^\dagger_{\vec{k}_n}|0\rangle. \tag{85}$$

These states represent n-particle states of the quantized field.

Conclusion: Quantizing the free scalar field follows the exact same steps as quantizing a harmonic oscillator. Each mode of the field behaves as an independent harmonic oscillator with its own frequency $\omega_{\vec{k}}$. The resulting quantum field is therefore an infinite collection of oscillators, and its excitations correspond to particles.

Lecture 6: Hamiltonian and Zero Point Energy in Free Scalar Field Theory

Last time, we quantized the free scalar field theory with the action

$$S = \int \mathrm{d}^4x \left(-\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 \right). \tag{86}$$

The conjugate momentum density and the Hamiltonian density are

$$\Pi = \partial_t\phi, \qquad \mathcal{H} = \frac{1}{2}\Pi^2 + \frac{1}{2}\left(\nabla\phi\right)^2 + \frac{1}{2}m^2\phi^2. \tag{87}$$

The classical equation of motion is

$$(\partial^2 - m^2)\phi = 0. \tag{88}$$

Following the general quantization procedure, we promote ϕ and Π to operators and obtain the complete solution to the operator equation:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{\omega_{\vec{k}}}}\left[\hat{a}_{\vec{k}}u_{\vec{k}} + \hat{a}^\dagger_{\vec{k}}u^*_{\vec{k}}\right], \tag{89}$$

where $u_{\vec{k}}$ is a complete set of mode functions:

$$u_{\vec{k}}(x) = e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}}, \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}. \tag{90}$$

The conjugate momentum density is

$$\hat{\Pi}(x) = \partial_t\hat{\phi}(x) = -i\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\sqrt{\omega_{\vec{k}}}\left[\hat{a}_{\vec{k}}u_{\vec{k}} - \hat{a}^\dagger_{\vec{k}}u^*_{\vec{k}}\right]. \tag{91}$$

Canonical Commutation Relations: The fundamental commutation relations are

$$[\hat{\phi}(t,\vec{x}), \hat{\phi}(t,\vec{x}')] = [\hat{\Pi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')] = 0, \qquad [\hat{\phi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'). \tag{92}$$

From these, one finds

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = [\hat{a}^\dagger_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = 0, \qquad [\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = (2\pi)^3\delta^{(3)}(\vec{k} - \vec{k}'). \tag{93}$$

Each mode $\vec{k}$ behaves as an independent harmonic oscillator. The commutator (93) is a continuum generalization of the familiar $[\hat{a}, \hat{a}^\dagger] = 1$ relation for a single oscillator. Different $\vec{k}$-modes commute because of the delta function.

Hamiltonian in Terms of $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}}$

The total Hamiltonian is obtained by integrating the Hamiltonian density:

$$H = \int \mathrm{d}^3x\,\mathcal{H} = \int \mathrm{d}^3x \left[\frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2\right]. \tag{94}$$

Substituting the field expansions, one finds

$$H = \frac{1}{2}\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\omega_{\vec{k}}\left(\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} + \hat{a}_{\vec{k}}\hat{a}^\dagger_{\vec{k}}\right) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\omega_{\vec{k}}\left(\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} + \frac{1}{2}[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}}]\right). \tag{95}$$

Using the commutation relation (93), we obtain

$$H = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\omega_{\vec{k}}\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} + E_0, \tag{96}$$

where the vacuum energy (zero-point energy) is

$$E_0 = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2}\omega_{\vec{k}}(2\pi)^3\delta^{(3)}_{(\vec{k})}(0). \tag{97}$$

Interpretation of $\delta^{(3)}_{(\vec{k})}(0)$: Recall the definition of the delta function:

$$(2\pi)^3\delta^{(3)}(\vec{k}) = \int \mathrm{d}^3x\, e^{i\vec{k}\cdot\vec{x}}. \tag{98}$$

Setting $\vec{k} = 0$, we find

$$(2\pi)^3\delta^{(3)}(0) = \int \mathrm{d}^3x = V, \tag{99}$$

which represents the spatial volume of the system. Hence, the zero-point energy can be written as

$$E_0 = V\,\epsilon_0, \qquad \epsilon_0 = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2}\omega_{\vec{k}}, \tag{100}$$

where ϵ_0 is the vacuum energy density.

Divergence of ϵ_0: Substituting $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$, we have

$$\epsilon_0 = \frac{1}{2}\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\sqrt{\vec{k}^2 + m^2}, \tag{101}$$

which is divergent.

Physically, this divergence arises because quantum field theory has a *continuum* of degrees of freedom. If we imagine a discrete lattice model with spacing a, each lattice point hosts a harmonic oscillator. In the continuum limit $a \to 0$, any finite volume contains infinitely many oscillators, leading to divergent total zero-point energy.

This is the first example of a divergence in QFT. Dealing with such divergences and understanding how they appear and how to remove or renormalize them is one of the major challenges and defining features of quantum field theory.

Physical Interpretation and Conserved Quantities

Time Independence of H: Even though $\hat{\phi}$ and $\hat{\Pi}$ are time-dependent operators, the Hamiltonian H itself is conserved because of time translation symmetry. The time integration is not performed in constructing H, only a spatial integration is done. The conservation of H follows from Noethers theorem applied to time translation symmetry.

Momentum Operator: Spatial translation symmetry gives rise to another conserved Noether charge:

$$P^i = -\int \mathrm{d}^3x\,\Pi(t,\vec{x})\,\partial_i\phi(t,\vec{x}). \tag{102}$$

In terms of the creation and annihilation operators, this becomes

$$P^i = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}k^i\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}}. \tag{103}$$

Unlike the Hamiltonian, there is no zero-point contribution in P^i, since the vacuum state carries zero momentum.

The theory also possesses Lorentz invariance, which leads to conserved angular momentum and boost charges $M^{\mu\nu}$. These too can be expressed in terms of $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}}$.

Regularization and the Cosmological Constant Problem

To make sense of the divergent zero-point energy, we introduce a momentum cutoff Λ:

$$\epsilon_0 = \frac{1}{2}\int^{\Lambda} \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\sqrt{\vec{k}^2 + m^2}, \tag{104}$$

which restricts the integral to momenta $|\vec{k}| < \Lambda$. This cutoff corresponds to the scale at which our effective field theory ceases to be valid. In a lattice model, $\Lambda \sim 1/a$.

Even with the cutoff, ϵ_0 remains a very large number. Physically, this zero-point energy is measurable in principle, but we do not observe such an enormous vacuum energy in the universe. This discrepancy between theoretical prediction and observation is known as the **cosmological constant problem**.

Remarks

- The Hamiltonian is time-independent due to time translation symmetry.
- The quantum field theory of ϕ reduces to a continuum of independent harmonic oscillators labeled by $\vec{k}$, each with frequency $\omega_{\vec{k}} = \sqrt{m^2 + \vec{k}^2}$.
- This picture connects naturally with a lattice of coupled oscillators: the continuum limit ($a \to 0$) yields infinitely many oscillators, one per point in space.
- The divergences encountered here are a direct consequence of the continuum nature of fields. Handling these divergences is central to modern quantum field theory.

Lecture 7. Hilbert Space and Particle Interpretation in Free Scalar QFT

In the free scalar field theory, each momentum mode behaves like an independent harmonic oscillator. This allows us to construct the Hilbert space of the theory in direct analogy with the harmonic oscillator in quantum mechanics.

Vacuum State and Fock-Space Construction

For each momentum mode $\vec{k}$, we introduce the operators $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}}$. The **vacuum state** is defined by

$$\hat{a}_{\vec{k}}\,|0\rangle = 0, \qquad \forall\,\vec{k}. \tag{105}$$

We postulate the existence of such a state. Later, one can explicitly write the wave functional of $|0\rangle$ in terms of $\phi(\vec{x})$, similar to the ground-state wave function of the harmonic oscillator.

Once the vacuum is defined, all states are obtained by acting with creation operators:

$$|n_{\vec{k}_1}, n_{\vec{k}_2}, \ldots\rangle \propto \left(\hat{a}^\dagger_{\vec{k}_1}\right)^{n_{\vec{k}_1}} \left(\hat{a}^\dagger_{\vec{k}_2}\right)^{n_{\vec{k}_2}} \cdots |0\rangle. \tag{106}$$

Thus the Hilbert space is the tensor product of infinitely many harmonic oscillators, one for each momentum $\vec{k}$.

Single- and Two-Particle States

The simplest excited state is

$$|\vec{k}\rangle = \hat{a}^\dagger_{\vec{k}}|0\rangle. \tag{107}$$

Two excitations give

$$|\vec{k}_1, \vec{k}_2\rangle = \hat{a}^\dagger_{\vec{k}_1}\,\hat{a}^\dagger_{\vec{k}_2}\,|0\rangle. \tag{108}$$

Because the creation operators commute,

$$[\hat{a}^\dagger_{\vec{k}_1}, \hat{a}^\dagger_{\vec{k}_2}] = 0,$$

all multi-particle states are symmetric under exchange. Therefore, the quanta of a real scalar field are **bosons**.

Energy–Momentum Eigenvalues

Ignoring the zero-point energy E_0, the vacuum satisfies

$$H|0\rangle = 0, \qquad P^i|0\rangle = 0. \tag{109}$$

One may show (exercise for students) that

$$H|\vec{k}\rangle = \omega_{\vec{k}}|\vec{k}\rangle, \qquad P^i|\vec{k}\rangle = k^i|\vec{k}\rangle, \tag{110}$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. Thus the one-particle state carries the 4-momentum

$$p^\mu = (\omega_{\vec{k}}, \vec{k}),$$

which satisfies $p^2 = -m^2$. Hence the excitation created by $\hat{a}^\dagger_{\vec{k}}$ is interpreted as a relativistic particle of mass m.

For the two-particle state,

$$H|\vec{k}_1, \vec{k}_2\rangle = (\omega_{\vec{k}_1} + \omega_{\vec{k}_2})|\vec{k}_1, \vec{k}_2\rangle, \tag{111}$$

$$\vec{P}|\vec{k}_1, \vec{k}_2\rangle = (\vec{k}_1 + \vec{k}_2)\,|\vec{k}_1, \vec{k}_2\rangle. \tag{112}$$

The energies simply add, reflecting the fact that the free theory has no interactions.

Normalization and Lorentz Covariance

The plane-wave states satisfy

$$\langle\vec{k}|\vec{k}'\rangle = (2\pi)^3\delta^{(3)}(\vec{k}-\vec{k}'). \tag{113}$$

This normalization is convenient but not Lorentz covariant. A better choice is the covariantly normalized state

$$|k\rangle = \sqrt{2\omega_{\vec{k}}}\,|\vec{k}\rangle, \tag{114}$$

for which

$$\langle k|k'\rangle = 2\omega_{\vec{k}}(2\pi)^3\delta^{(3)}(\vec{k}-\vec{k}'). \tag{115}$$

The factor $\omega_{\vec{k}}\delta^{(3)}(\vec{k}-\vec{k}')$ transforms simply under Lorentz transformations.

If U_Λ denotes the unitary operator associated with a Lorentz transformation Λ, then

$$U_\Lambda|k\rangle = |\Lambda k\rangle. \tag{116}$$

Physical Wave Packets

The states $|\vec{k}\rangle$ and $|k\rangle$ are not normalizable, just like plane waves $e^{i\vec{k}\cdot\vec{x}}$ in non-relativistic quantum mechanics. A **physical** one-particle state is a wave packet,

$$|\Psi\rangle = \int \frac{d^3\vec{k}}{(2\pi)^3}\, f(\vec{k})\,|k\rangle, \tag{117}$$

with normalization

$$\langle\Psi|\Psi\rangle = \int \frac{d^3\vec{k}}{(2\pi)^3}|f(\vec{k})|^2 < \infty. \tag{118}$$

Choosing $f(\vec{k})$ appropriately, one can construct localized wave packets. Similarly, a two-particle wave packet takes the form

$$\int \frac{d^3\vec{k}_1}{(2\pi)^3}\frac{d^3\vec{k}_2}{(2\pi)^3} f(\vec{k}_1,\vec{k}_2)\,|k_1,k_2\rangle. \tag{119}$$

Structure of the Hilbert Space: Fock Space

Because the theory is free, the Hilbert space factorizes as

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \cdots, \tag{120}$$

where $\mathcal{H}_0$ is the vacuum sector, $\mathcal{H}_1$ the one-particle sector, $\mathcal{H}_2$ the two-particle sector, and so on. The direct sum of all sectors with finite particle number is the **Fock space**.

Remarks

- Each momentum mode is a harmonic oscillator. Each excitation corresponds to a particle.
- Since $[\hat{a}^\dagger_{\vec{k}_1}, \hat{a}^\dagger_{\vec{k}_2}] = 0$, all particles are bosons.
- All physical energies are positive, despite $E^2 = \vec{k}^2 + m^2$ admitting $\pm\omega_{\vec{k}}$.
- In the free theory, the total energy is the sum of the individual energies; there is no interaction energy.
- Plane-wave states are not normalizable, so physical states must be constructed as wave packets.
- Although $\vec{k}$ labels a continuum, the space of normalizable one-particle states is separable and admits a countable orthonormal basis, just as in ordinary quantum mechanics.

Lecture 8 Conserved Charges, Noether's Theorem, and Complex Scalar Fields

In the previous lectures we quantized the free scalar field and interpreted its excitations as relativistic particles. We also discussed the structure of the Hilbert space and the role of conserved quantities such as energy and momentum. In this lecture we study Noether's theorem in field theory, the quantum meaning of conserved charges, and the appearance of particles and anti-particles in the complex scalar field.

Noether's Theorem and Conserved Currents

Consider a set of fields ϕ_a and a continuous symmetry acting on them:

$$\phi_a \;\longrightarrow\; \phi'_a = \phi_a + \epsilon_\alpha \, f_a{}^\alpha(\phi, \partial\phi). \tag{121}$$

Here:

- ϵ_α are infinitesimal parameters, one for each symmetry,
- a labels the fields, and α labels the symmetries.

If the action is invariant under this transformation up to a total derivative,

$$\delta\mathcal{L} = \epsilon_\alpha \partial_\mu K^\mu{}_\alpha, \tag{122}$$

then Noether's theorem tells us that the current

$$J^\mu{}_\alpha = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} \, f_a{}^\alpha - K^\mu{}_\alpha \tag{123}$$

is conserved:

$$\partial_\mu J^\mu{}_\alpha = 0. \tag{124}$$

The associated conserved charge is

$$Q_\alpha = \int d^3x \, J^0{}_\alpha. \tag{125}$$

The index position on α is irrelevant; it simply labels the symmetry.

Quantum Interpretation: Charges Generate Symmetries

At the quantum level the fields become operators $\hat\phi_a$, the current becomes $\hat J^\mu{}_\alpha$, and the conserved charge becomes an operator

$$\hat Q_\alpha = \int d^3x \, \hat J^0{}_\alpha.$$

Since $\partial_\mu J^\mu_\alpha = 0$, the quantum charge is time independent:

$$\frac{d\hat Q_\alpha}{dt} = 0.$$

The fundamental result is that the operator $\hat Q_\alpha$ generates the symmetry. For infinitesimal transformations,

$$[\, \epsilon_\alpha \hat Q_\alpha, \; \hat\phi_a(x) \,] = -i\epsilon_\alpha f_a{}^\alpha(x). \tag{126}$$

Simplified Proof for Internal Symmetries

For internal symmetries, f_a^α typically contains no time derivatives and $K^0_\alpha = 0$. In this case,

$$\hat J^0_\alpha = \hat\Pi_a f_a^\alpha,$$

where

$$\Pi_a = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi_a)} \quad \longrightarrow \quad \text{operator } \hat\Pi_a$$

after quantization. Using the equal-time commutation relation

$$[\, \hat\phi_a(t, \vec x), \hat\Pi_b(t, \vec x\,') \,] = i\delta_{ab} \, \delta^{(3)}(\vec x - \vec x\,'), \tag{127}$$

one obtains

$$[\, \epsilon_\alpha \hat Q_\alpha, \; \hat\phi_a(t, \vec x) \,] = -i\epsilon_\alpha f_a^\alpha.$$

Example: Time Translations A useful example is the symmetry under time translations. Recall that the Hamiltonian density is defined as

$$\hat{\mathcal{H}} = \hat{\Pi}_a \, \partial_t \hat{\phi}_a - \mathcal{L}. \tag{128}$$

For a time translation, the change of the Lagrangian is a total derivative, so $\mathcal{L}$ plays the role of the quantity $K^\mu{}_\alpha$ in $\delta\mathcal{L} = \epsilon_\alpha \partial_\mu K^\mu{}_\alpha$. The corresponding conserved charge is the Hamiltonian operator

$$\hat{H} = \int d^3x \, \hat{\mathcal{H}}. \tag{129}$$

Using the equal-time commutation relations, we find

$$[\,\hat{H},\, \hat{\phi}_a(t, \vec{x})\,] = -\,i\,\partial_t \hat{\phi}_a(t, \vec{x}), \tag{130}$$

which is exactly the Heisenberg equation of motion. This shows that $\hat{H}$ acts as the generator of time translations. Infinitesimally, a time translation changes the field by

$$\delta\hat{\phi}_a = \epsilon \, \partial_t \hat{\phi}_a,$$

and this transformation is precisely generated by the commutator with $\hat{H}$.

Finite symmetry transformation

A finite symmetry transformation is generated by the unitary operator

$$\hat{U}(\Lambda_\alpha) = \exp\Big(i\Lambda_\alpha \hat{Q}_\alpha\Big)\,, \tag{131}$$

which acts on the field operator as

$$\hat{U}(\Lambda_\alpha)\,\hat{\phi}_a(x)\,\hat{U}^\dagger(\Lambda_\alpha) = \hat{\phi}'_a(x). \tag{132}$$

Expanding $\hat{U}$ to first order in Λ_α reproduces the infinitesimal result (126).

Example: Time Translations The finite time translation is produced by the unitary operator

$$\hat{U}(t) = e^{i\hat{H}t}, \tag{133}$$

and we have

$$\hat{U}(t)\,\hat{\phi}(0, \vec{x})\,\hat{U}^\dagger(t) = \hat{\phi}(t, \vec{x}), \tag{134}$$

which is the operator statement that $\hat{H}$ generates time evolution.

Complex Scalar Field and Anti-Particles

Consider the complex scalar Lagrangian

$$\mathcal{L} = -\partial_\mu \phi \, \partial^\mu \phi^\dagger - m^2 \phi^\dagger \phi. \tag{135}$$

The momenta are

$$\hat{\Pi}_\phi = \partial_0 \hat{\phi}^\dagger, \qquad \hat{\Pi}_{\phi^\dagger} = \partial_0 \hat{\phi}.$$

The mode expansion of the quantum field is

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\hat{a}_{\vec{k}} \, e^{-i\omega_k t + i\vec{k}\cdot\vec{x}} + \hat{b}^\dagger_{\vec{k}} \, e^{i\omega_k t - i\vec{k}\cdot\vec{x}} \right], \tag{136}$$

with

$$[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \qquad [\hat{b}_{\vec{k}}, \hat{b}^\dagger_{\vec{k}'}] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \tag{137}$$

The vacuum satisfies

$$\hat{a}_{\vec{k}}|0\rangle = \hat{b}_{\vec{k}}|0\rangle = 0.$$

Thus there are two sets of quanta:

$$\hat{a}^\dagger_{\vec{k}}|0\rangle, \qquad \hat{b}^\dagger_{\vec{k}}|0\rangle,$$

both with mass m.

The U(1) Charge and Particle, Anti-Particle Interpretation

The Lagrangian is invariant under the phase rotation

$$\phi \to e^{i\alpha}\phi.$$

The Noether charge operator is

$$\hat{Q} = i \int d^3x \left(\hat{\Pi}_{\phi^\dagger}\hat{\phi}^\dagger - \hat{\Pi}_\phi \hat{\phi}\right). \tag{138}$$

or

$$\hat{Q} = i \int d^3x \left(\hat{\phi}^\dagger \hat{\Pi}_{\phi^\dagger} - \hat{\phi}\hat{\Pi}_\phi\right). \tag{139}$$

which has the ordering ambiguity. To resolve ordering ambiguities, we impose

$$\hat{Q}|0\rangle = 0.$$

One then obtains

$$\hat{Q} = \int \frac{d^3k}{(2\pi)^3} \left(\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} - \hat{b}^\dagger_{\vec{k}}\hat{b}_{\vec{k}}\right). \tag{140}$$

Thus:

$$[\hat{Q}, \hat{a}^\dagger_{\vec{k}}] = +\hat{a}^\dagger_{\vec{k}}, \tag{141}$$

$$[\hat{Q}, \hat{b}^\dagger_{\vec{k}}] = -\hat{b}^\dagger_{\vec{k}}. \tag{142}$$

Define covariantly normalized states:

$$|k\rangle = \sqrt{2\omega_k}\,\hat{a}^\dagger_{\vec{k}}|0\rangle, \qquad |\bar{k}\rangle = \sqrt{2\omega_k}\,\hat{b}^\dagger_{\vec{k}}|0\rangle.$$

Then

$$\hat{Q}|k\rangle = +|k\rangle, \qquad \hat{Q}|\bar{k}\rangle = -|\bar{k}\rangle.$$

Thus:

- The $\hat{a}^\dagger$ quanta carry charge $+1$,
- The $\hat{b}^\dagger$ quanta carry charge -1,
- They have the same mass and spin zero.

These are interpreted as a **particle** and its **anti-particle**.

Remark. In relativistic QFT every particle has an associated anti-particle. For a real scalar field, $\hat{\phi}^\dagger = \hat{\phi}$, so the particle is its own anti-particle.

Lecture 9: Propagators in Quantum Mechanics and Quantum Field Theory

1. Propagators in Non-Relativistic Quantum Mechanics

In non-relativistic quantum mechanics, the position eigenstate $|\vec{x}\rangle$ satisfies

$$\hat{\vec{x}}\,|\vec{x}\rangle = \vec{x}\,|\vec{x}\rangle, \tag{143}$$

and the wave function is defined as

$$\psi(\vec{x}) = \langle\vec{x}|\psi\rangle. \tag{144}$$

In the Heisenberg picture, the position operator becomes time-dependent:

$$\hat{\vec{x}}(t)\,|\vec{x},t\rangle = \vec{x}\,|\vec{x},t\rangle. \tag{145}$$

The propagator is defined as

$$G(\vec{x},t;\vec{x}',t') \equiv \langle\vec{x},t|\vec{x}',t'\rangle. \tag{146}$$

If the wave function is known at time t', then:

$$\psi(t,\vec{x}) = \int d^3x'\,G(\vec{x},t;\vec{x}',t')\,\psi(t',\vec{x}'). \tag{147}$$

The propagator completely determines the quantum dynamics.

2. Localized States and Their Problems in Relativistic QFT

In relativistic QFT, one might attempt to generalize $|\vec{x},t\rangle$ to define a relativistic propagator, but this fails.

Why localized states fail

- In QFT, there is no position operator. The coordinate x is only a label, not an operator.
- A truly localized state would satisfy

  $$\langle\vec{x},t|\vec{x}',t\rangle = \delta^{(3)}(\vec{x}-\vec{x}'), \tag{148}$$

  but $\delta^{(3)}(\vec{x}-\vec{x}')$ is not Lorentz covariant.
- Therefore one finds

  $$U_\Lambda|\vec{x},t\rangle \neq |\Lambda x\rangle, \tag{149}$$

  so Lorentz symmetry is broken.

Thus, QFT cannot use localized position eigenstates.

3. The Closest Analogue: Field-Excited States

The best Lorentz-covariant analogue of a position eigenstate is:

$$|x\rangle \equiv \hat{\phi}(x)\,|0\rangle, \qquad \langle x| \equiv \langle 0|\,\hat{\phi}(x), \tag{150}$$

where $\hat{\phi}(x)$ is a real scalar field operator, thus it is Hermitian operator $\hat{\phi} = \hat{\phi}^\dagger$.

Under Lorentz transformation:

$$U_\Lambda|x\rangle = |\Lambda x\rangle. \tag{151}$$

The inner product is:

$$\langle x|x'\rangle = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle. \tag{152}$$

This is not a delta function, so $|x\rangle$ is not localized, but it is Lorentz covariant.

4. “Wave Functions in QFT

Given a single-particle state $|\psi\rangle$, define:

$$\psi(x) \equiv \langle x|\psi\rangle = \langle 0|\, \hat{\phi}(x)\, |\psi\rangle. \tag{153}$$

If $|\psi\rangle = |k\rangle$ is a momentum eigenstate:

$$\langle x|k\rangle = \langle 0|\hat{\phi}(x)|k\rangle = e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}}. \tag{154}$$

Although this looks like a wave function, $|\psi(x)|^2$ is not a position probability density because $|x\rangle$ is not a true position eigenstate.

5. Wightman Functions and Propagators in QFT

The most important QFT two-point function is the Wightman function:

$$G_+(x, x') \equiv \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle = \langle x|x'\rangle. \tag{155}$$

The reversed ordering gives:

$$G_-(x, x') \equiv \langle 0|\hat{\phi}(x')\hat{\phi}(x)|0\rangle. \tag{156}$$

Retarded, Advanced, and Feynman Propagators

$$G_R(x, x') = \theta(t - t')\, \langle 0|[\hat{\phi}(x), \hat{\phi}(x')]|0\rangle, \tag{157}$$

$$G_A(x, x') = -\theta(t' - t)\, \langle 0|[\hat{\phi}(x), \hat{\phi}(x')]|0\rangle, \tag{158}$$

$$G_F(x, x') = \theta(t - t')\, G_+(x, x') + \theta(t' - t)\, G_-(x, x'). \tag{159}$$

where G_R is the retarded propagator, G_A is the advanced propagator, and G_F is the Feynman propagator.

6. Equations of Motion of Propagators

The field satisfies the KleinGordon equation:

$$(\partial^2 + m^2)\hat{\phi}(x) = 0.$$

Thus,

$$(\partial^2 + m^2)G_\pm(x, x') = 0. \tag{160}$$

For G_R, G_A, and G_F, derivatives of θ produce delta-functions:

$$(\partial^2 + m^2)G_{R,A,F}(x, x') = -i\, \delta^{(4)}(x - x'). \tag{161}$$

Hence these propagators are Green’s functions of the KG operator.

7. Microcausality

The key commutator is:

$$\hat{\Delta}(x, x') \equiv [\hat{\phi}(x), \hat{\phi}(x')]. \tag{162}$$

For spacelike separations:

$$[\hat{\phi}(x), \hat{\phi}(x')] = 0, \qquad (x - x')^2 > 0. \tag{163}$$

This ensures relativistic causality. Thus:

$$G_+ = G_- = G_F, \qquad G_R = G_A = \Delta = 0, \qquad (x - x')^2 > 0.$$

8. Symmetry Properties

The vacuum is invariant under translations and Lorentz transformations, so:

$$G_{R,A,F,\pm}(x, x') = G_{R,A,F,\pm}(x - x') = G_{R,A,F,\pm}((x - x')^2). \tag{164}$$

Thus the propagators depend only on the invariant separation.

9. Summary

- Non-relativistic propagators evolve wave functions between times.
- QFT cannot use localized position eigenstates due to Lorentz symmetry.
- The closest replacement is $|x\rangle = \hat{\phi}(x)|0\rangle$.
- Two-point functions such as G_+ and G_F encode the structure of the theory.
- $G_{R,A,F}$ satisfy Klein-Gordon Green's equations with a delta-function source.
- Microcausality requires $[\hat{\phi}(x), \hat{\phi}(x')] = 0$ for spacelike separation.
- All propagators depend only on the invariant $x - x'$ because of symmetries.

Lecture 10: Computation of Propagators

1. Computing the Wightman Function $G_+(x,x')$

We begin with the definition

$$G_+(x,x') = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle. \tag{165}$$

Using the mode expansion

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\left(\hat{a}_{\vec{k}}\, u_{\vec{k}}(x) + \hat{a}^\dagger_{\vec{k}}\, u^*_{\vec{k}}(x)\right),$$

we obtain

$$G_+(x,x') = \int \frac{d^3k}{(2\pi)^3}\frac{d^3k'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\frac{1}{\sqrt{2\omega_{\vec{k}'}}}\langle 0|\left(\hat{a}_{\vec{k}}u_{\vec{k}}(x) + \hat{a}^\dagger_{\vec{k}}u^*_{\vec{k}}(x)\right)\left(\hat{a}_{\vec{k}'}u_{\vec{k}'}(x') + \hat{a}^\dagger_{\vec{k}'}u^*_{\vec{k}'}(x')\right)|0\rangle. \tag{166}$$

The only non-vanishing contribution comes from $\hat{a}_{\vec{k}}\hat{a}^\dagger_{\vec{k}'}$:

$$G_+(x,x') = \int \frac{d^3k}{(2\pi)^3}\frac{1}{2\omega_{\vec{k}}}e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')}. \tag{167}$$

This formula is correct but not manifestly Lorentz covariant. We now show how to rewrite it in a covariant 4-dimensional form.

Step 1: Write the exponent covariantly. With the metric $(-,+,+,+)$,

$$k\cdot(x-x') = -k^0(t-t') + \vec{k}\cdot(\vec{x}-\vec{x}').$$

On shell, $k^0 = \omega_{\vec{k}} > 0$, so

$$e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')} = e^{\,ik\cdot(x-x')}.$$

Step 2: Promote the 3D integral to a 4D integral. Insert the identity

$$1 = \int \frac{dk^0}{2\pi}\, 2\pi\,\delta(k^0-\omega_{\vec{k}}).$$

Then

$$G_+(x,x') = \int \frac{d^4k}{(2\pi)^4}\frac{2\pi}{2\omega_{\vec{k}}}\delta(k^0-\omega_{\vec{k}})e^{\,ik\cdot(x-x')}.$$

Step 3: Replace the delta function with a covariant expression. Using

$$\delta(k^2+m^2) = \frac{1}{2\omega_{\vec{k}}}\left[\delta(k^0-\omega_{\vec{k}}) + \delta(k^0+\omega_{\vec{k}})\right],$$

and keeping only positive-energy modes gives

$$\theta(k^0)\,\delta(k^2+m^2) = \frac{1}{2\omega_{\vec{k}}}\delta(k^0-\omega_{\vec{k}}).$$

Step 4: Substitute back. We obtain the manifestly Lorentz-covariant expression

$$G_+(x,x') = \int \frac{d^4k}{(2\pi)^4}\, 2\pi\,\theta(k^0)\,\delta(k^2+m^2)\,e^{\,ik\cdot(x-x')}, \tag{168}$$

which is the covariant representation of the Wightman function G_+.

2. Behavior at Equal Times

At $t = t'$,

$$G_+(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{i\vec{k}\cdot(\vec{x}-\vec{x}')} \neq \delta^{(3)}(\vec{x} - \vec{x}'). \tag{169}$$

Let $r = |\vec{x} - \vec{x}'|$. The integral is:

$$G_+(r) \propto K_0(mr),$$

where K_0 is a modified Bessel function. For $mr \gg 1$,

$$G_+(r) \sim e^{-mr}.$$

Thus the correlation length is

$$\xi = \frac{1}{m}.$$

If two points are separated by more than a correlation wavelength, their field correlation becomes exponentially small.

Fig: Exponential decay of $G_+(r)$ with correlation length $\xi = 1/m$.

3. Fourier-Space Representation

Fourier transform convention:

$$G(k) = \int d^4x\, G(x)\, e^{-ik\cdot x}, \qquad G(x) = \int \frac{d^4k}{(2\pi)^4} G(k) e^{ik\cdot x}. \tag{170}$$

Comparing with (168), we obtain:

$$G_+(k) = 2\pi\, \theta(k^0)\delta(k^2 + m^2). \tag{171}$$

4. General Green's Function via Fourier Transform

Starting from the Klein–Gordon equation,

$$(\partial^2 + m^2)\, G_{R,A,F}(x - x') = -i\delta^{(4)}(x - x'),$$

its Fourier transform satisfies

$$(k^2 + m^2)\, G(k) = -i \qquad \Longrightarrow \qquad G(k) = \frac{-i}{k^2 + m^2}.$$

Thus the coordinate-space Green's function becomes

$$G(x) = \int \frac{d^4k}{(2\pi)^4} \frac{-i\, e^{ik\cdot x}}{k^2 + m^2}. \tag{172}$$

The denominator contains

$$k^2 + m^2 = -\omega^2 + \vec{k}^2 + m^2 = -\omega^2 + \omega_{\vec{k}}^2, \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2},$$

so the integrand has two poles on the real axis:

$$\omega = \pm\omega_{\vec{k}}.$$

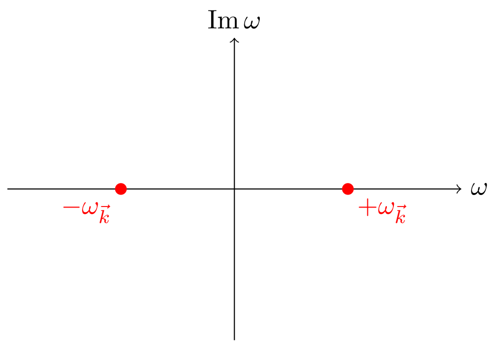


FIG: The two poles of the integrand on the real axis.

Because these poles lie exactly on the integration contour (the real axis), the integral is not well-defined. To evaluate the integral, we must choose a contour that avoids the poles and specifies how the integral is to be interpreted. Different contour choices correspond to different causal propagators.

5. Contour Choices and Causal Propagators

By Cauchys theorem,

$$\oint_C f(z)\,dz = 2\pi i \sum_{\text{poles inside } C} \mathrm{Res}(f),$$

so:

- If the closed contour encloses no poles, the integral is 0.
- If it encloses one or more poles, the value is $2\pi i$ times the sum of their residues.

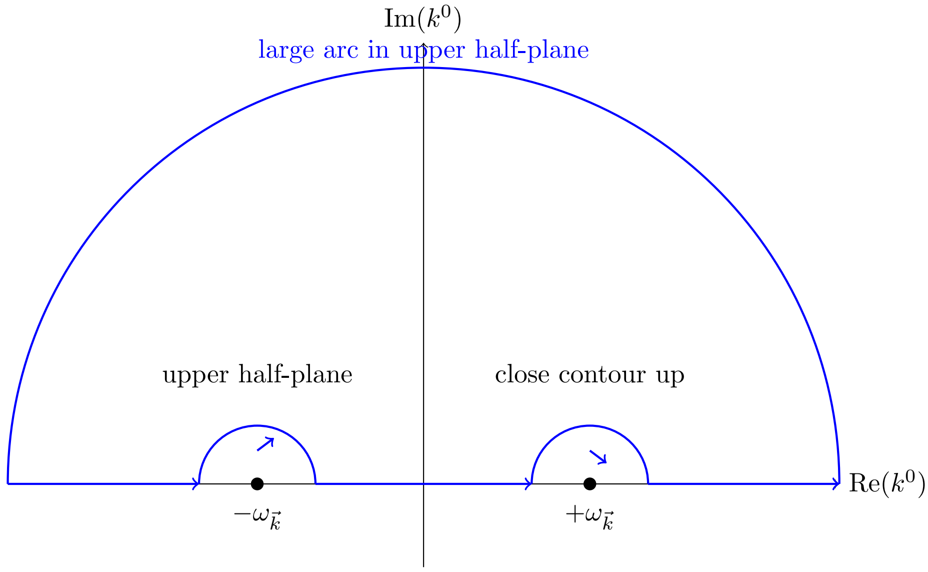
Thus, to obtain the retarded, advanced, and Feynman propagators, we choose contours that enclose the correct poles consistent with the desired causal structure. Below we explain each case clearly.

- **Retarded propagator G_R: close the contour in the *upper half-plane.***

  Physical condition:

  $$G_R(x - x') = 0 \qquad \text{for } t < t'.$$

  To achieve this, for $t < t'$, the factor $e^{-ik^0(t-t')}$ should decay in the upper half-plane, so we close the contour upward. Since all poles lie below, no poles are enclosed, and the integral is 0 for $t < t'$. This produces a propagator that only responds to sources in the past, therefore it's called "retarded".

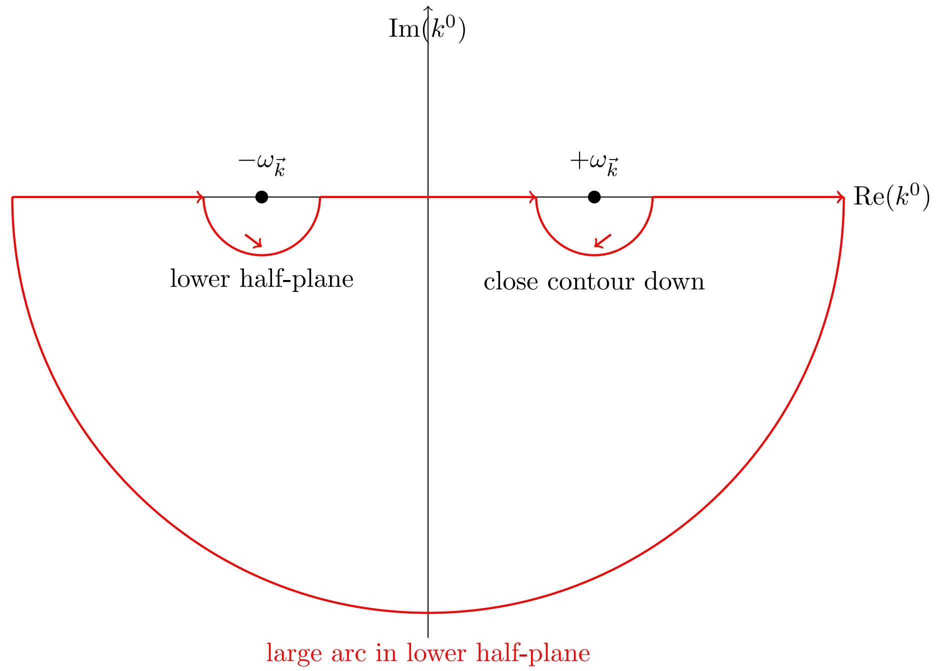


- **Advanced propagator G_A: close the contour in the *lower half-plane.***

  Physical condition:

  $$G_A(x - x') = 0 \qquad \text{for } t > t'.$$

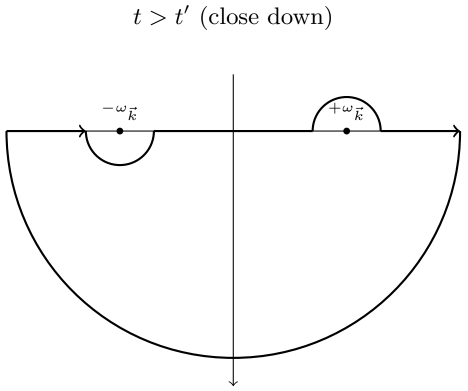
  To achieve this, we close the contour downward encloses no poles, so $G_A = 0$ when $t > t'$. This propagator only responds to sources in the *future*, hence it is"advanced".

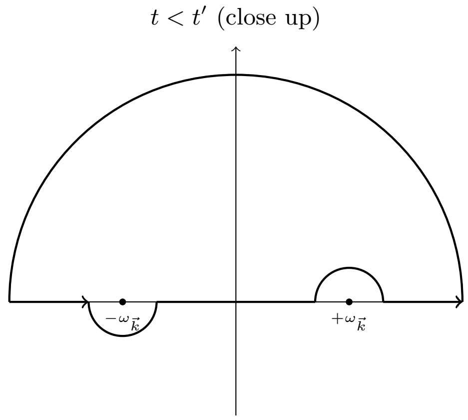


- **Feynman propagator G_F: one pole above and one pole below.**
  Goal: recover the time-ordered propagator:

$$G_F(x, x') = \theta(t - t')G_+(x, x') + \theta(t' - t)G_-(x, x').$$

  To accomplish this: For $t > t'$, we close the contour downward; For $t < t'$, we close the contour upward. So the Feynman propagator always takes the contribution from the pole consistent with time ordering.





6. Summary

- $G_+(x, x')$ can be computed exactly and decays exponentially with distance.
- Correlation length: $\xi = 1/m$.
- Different causal propagators arise from different contour deformations around poles.

Lecture 11: Propagators, Contours, Interacting Fields, and Scattering

1. Review of the Propagators

For a real scalar field, the Wightman functions are

$$G_+(x,x') = \langle 0|\phi(x)\phi(x')|0\rangle, \qquad G_-(x,x') = \langle 0|\phi(x')\phi(x)|0\rangle. \tag{173}$$

They admit two interpretations:

- **Heuristic particle picture:** $G_+(x,x')$ denotes the transition amplitude for a particle to propagate from x' to x. This is only heuristic because $|x'\rangle$ is not a perfectly localized one-particle state.
- **Correlation function picture:** In condensed matter language, ϕ may represent spins or an order parameter, so $G_+(x,x')$ measures correlations of the field.

The **Feynman propagator** is the time-ordered combination

$$G_F(x,x') = \theta(t-t')\, G_+(x,x') + \theta(t'-t)\, G_-(x,x'). \tag{174}$$

It always propagates from the earlier to the later time, hence the name *time-ordered* two-point function.

Propagators from Fourier Transform and Contours

The general momentum-space expression is

$$G(x-x') = -i\int \frac{d^4k}{(2\pi)^4}\frac{e^{ik(x-x')}}{k^2+m^2}. \tag{175}$$

The poles satisfy

$$k^2+m^2 = -\omega^2+\omega_{\vec{k}}^2 = 0, \quad \Rightarrow \quad \omega = \pm\omega_{\vec{k}}.$$

Different choices of contour give different propagators:

- **Retarded:** contour passes above both poles $\Rightarrow G_R \propto \theta(t-t')$.
- **Advanced:** contour passes below both poles $\Rightarrow G_A \propto \theta(t'-t)$.
- **Feynman:** contour passes below the negative-energy pole and above the positive-energy pole.

The ϵ-prescription. Instead of redrawing contours, shift the poles:

$$G_{R,A}(x-x') = \int \frac{d^4k}{(2\pi)^4}\frac{-i\,e^{ik(x-x')}}{-(\omega\pm i\epsilon)^2+\omega_{\vec{k}}^2}, \tag{176}$$

$$G_F(x-x') = \int \frac{d^4k}{(2\pi)^4}\frac{-i\,e^{ik(x-x')}}{k^2+m^2-i\epsilon}. \tag{177}$$

n-Point Correlation Functions and Wick's Theorem

General correlators in any state $|\psi\rangle$:

$$\langle\psi|\phi(x_1)\phi(x_2)\cdots\phi(x_n)|\psi\rangle. \tag{178}$$

For the vacuum of a free theory, Wick's theorem gives

$$\langle 0|\phi(x_1)\cdots\phi(x_n)|0\rangle = \sum_{\text{pairings}} G_+(x_i,x_j). \tag{179}$$

Example: the 4-point function

$$\langle 0|\phi_1\phi_2\phi_3\phi_4|0\rangle = G_+(x_1,x_2)G_+(x_3,x_4) + G_+(x_1,x_4)G_+(x_2,x_3) + G_+(x_1,x_3)G_+(x_2,x_4). \tag{180}$$

Composite operators. Operators like $\phi^2(x)$, $\mathcal{H}(x)$, $T^{\mu\nu}(x)$ involve products of fields at the same point, which are typically singular. They must be renormalized to become well-defined.

2. Scattering and the S-Matrix

Figuring out interactions from experiments is one of the main tasks of particle physics and also of many areas in condensed matter physics.

In high-energy physics, we build large colliders such as the LHC at CERN and collide protons at extremely high energies. The entire purpose of these billion-dollar machines is to probe and understand the interactions between elementary particles, and to test and verify our theoretical models. In condensed matter physics, many experiments play a similar role on very different energy scales.

Scattering as the universal probe.

A natural question is: *What is the most powerful experimental way to probe interactions between particles?* The answer is: **scattering**. This idea has been used for about 100 years, starting from Rutherford, who shot alpha particles at atoms and deduced the existence of the atomic nucleus from the scattering pattern. Later, deep inelastic scattering (DIS) experiments revealed that the proton has substructure, leading to the discovery of quarks. Many particles and many kinds of interactions have been discovered using scattering.

The basic idea of a scattering experiment is simple:

- Prepare some incoming particles.
- Let them collide and interact.
- Measure the outgoing particles and their momenta.

From the observed outcomes, we deduce the nature of the interactions. In condensed matter, similar logic appears in neutron scattering, X-ray scattering, and related techniques where we "shine" probes on a sample and study the scattered waves. One of the central theoretical objects describing scattering processes is the **S-matrix**.

Idealized Definition of the S-Matrix

To define the S-matrix precisely, we use a standard mathematical idealization.

Asymptotic states. At very early times $t \to -\infty$, we prepare an initial state as a collection of localized wave packets that are spatially very far apart. We aim them so that they will eventually move toward one another, meet in some region, and scatter.

After they interact in some finite time window, the outgoing products fly away. At very late times $t \to +\infty$, these final particles are again far apart from each other. Because they are separated, their mutual interactions can be neglected and each particle can be cleanly identified.

This asymptotic separation is important:

- As $t \to \pm\infty$ we can treat the particles as *free*, so free field theory describes both the initial and final states.
- In contrast, during the actual collision, interactions are important and the full interacting theory is needed.

Thus the initial and final states are well-described as collections of free particles, even though the dynamics in between is governed by an interacting theory. This is crucial, since we cannot solve the interacting theory exactly, but we can describe the asymptotic free states with essentially arbitrary precision.

Labeling the initial and final states. For a scalar field theory (no spin), the only quantum numbers are the momenta. We denote the set of incoming momenta by

$$\alpha = \{p_1, \ldots, p_k\}, \tag{181}$$

which characterizes the initial state. Similarly, the set of outgoing momenta is

$$\beta = \{p_1', \ldots, p_n'\}, \tag{182}$$

and here n does not have to equal k. In general,

$$\alpha \to \beta$$

describes a scattering process in which k incoming particles with momenta $\{p_i\}$ scatter into n outgoing particles with momenta $\{p_j'\}$.

Definition and Basic Properties of the S-Matrix

In the Schrödinger picture, the time evolution operator $U(t_2, t_1)$ evolves states from t_1 to t_2. The amplitude for an initial state $|\alpha\rangle$ at $t = -\infty$ to become a final state $|\beta\rangle$ at $t = +\infty$ is

$$\langle\beta|U(+\infty, -\infty)|\alpha\rangle. \tag{183}$$

In the Heisenberg picture, this is often written as

$$\langle\beta, +\infty|\alpha, -\infty\rangle. \tag{184}$$

We denote this transition amplitude by

$$S_{\beta\alpha} \equiv \langle\beta, +\infty|\alpha, -\infty\rangle. \tag{185}$$

If we let α and β run over *all* possible choices of initial and final asymptotic states, the collection of $S_{\beta\alpha}$ forms an (infinite-dimensional) matrix called the **S-matrix**.

Free theory and interactions. The quantity $S_{\beta\alpha}$ is just a matrix element of the full evolution operator $U(+\infty, -\infty)$. If the interaction is very weak, most of the time the particles do not scatter, so we expect that

$$S \approx I \tag{186}$$

in the limit of no interactions. For a strictly free theory, S is exactly the identity matrix.

When interactions are weak but nonzero, it is convenient to write

$$S = I + iT, \qquad S_{\beta\alpha} = \delta_{\beta\alpha} + iT_{\beta\alpha}, \tag{187}$$

where T encodes the nontrivial interaction effects, and the factor of i is a conventional choice.

Momentum Conservation and the Scattering Amplitude

Because the underlying theory is invariant under spacetime translations, total energy-momentum is conserved in any scattering process. This implies that the interaction part $iT_{\beta\alpha}$ can be nonzero only if the total initial and total final momenta are equal:

$$\sum_{i=1}^{k} p_i = \sum_{j=1}^{n} p'_j. \tag{188}$$

Therefore, $iT_{\beta\alpha}$ must contain a delta function enforcing this conservation law. It is convenient to write

$$iT_{\beta\alpha} = i(2\pi)^4\delta^{(4)}(p_\alpha - p_\beta)\, M_{\alpha\to\beta}, \tag{189}$$

where

$$p_\alpha = \sum_{i=1}^{k} p_i, \qquad p_\beta = \sum_{j=1}^{n} p'_j$$

are the total 4-momenta of the initial and final states. The factor

$$M_{\alpha\to\beta} \tag{190}$$

is called the **scattering amplitude**. The delta function is the trivial kinematic part; all the interesting dynamical information about the interactions is contained in $M_{\alpha\to\beta}$.

Symmetries and Unitarity

The S-matrix is built from the full time evolution operator U, which is unitary. Therefore,

$$S \text{ is a unitary matrix.}$$

This leads to important constraints on the amplitudes $M_{\alpha\to\beta}$ (optical theorem, etc.), although we will not derive them here.

More generally, suppose there is a unitary operator Λ that generates a symmetry of the Hamiltonian:

$$[\Lambda, H] = 0, \qquad \Lambda\Lambda^\dagger = I. \tag{191}$$

Typical examples include Lorentz transformations, which we can schematically write as

$$\Lambda = e^{i\omega M}, \tag{192}$$

where M is the generator of the symmetry and ω the parameter.

Then the S-matrix is invariant under the symmetry:

$$S_{\Lambda\beta,\Lambda\alpha} = S_{\beta\alpha}. \tag{193}$$

This expresses the fact that if we transform both the initial and final states by a symmetry of the Hamiltonian, the scattering amplitude does not change.

The quantity $M_{\alpha\to\beta}$ is the central object in interacting quantum field theory. Later, we will see (via the LSZ reduction formula) that $M_{\alpha\to\beta}$ can be computed from time-ordered correlation functions of the interacting field.

Lecture 12: Interacting Theories, the S-Matrix, and Path Integrals

1. Interacting Scalar Field Theory

The simplest interacting relativistic field theory is the ϕ^4 theory with Lagrangian density

$$\mathcal{L} = -\partial_\mu \phi \, \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4. \tag{194}$$

We will use this model to illustrate how interactions affect particle dynamics and how scattering amplitudes are computed.

The key observable in an interacting QFT is the S**-matrix**. In the Schrödinger picture:

$$S_{\beta\alpha} = \langle \beta | U(+\infty, -\infty) | \alpha \rangle. \tag{195}$$

Here the initial state $|\alpha\rangle$ consists of widely separated incoming particles at $t \to -\infty$, which can be treated as free particles. As time evolves, they approach each other, scatter, and then separate again at very late times. The outgoing free particles at $t \to +\infty$ form the final state $|\beta\rangle$.

In the Heisenberg picture, the same amplitude is written as

$$S_{\beta\alpha} = \langle \beta, +\infty \,|\, \alpha, -\infty \rangle. \tag{196}$$

It is convenient to decompose S into

$$S_{\beta\alpha} = \delta_{\beta\alpha} + i \, T_{\beta\alpha}, \tag{197}$$

where the term $iT_{\beta\alpha}$ captures all effects of the interactions.

Because we always consider theories that are invariant under spacetime translations, total energy and momentum are conserved. Thus,

$$iT_{\beta\alpha} = i(2\pi)^4 \delta^{(4)}(p_\alpha - p_\beta) \, M_{\alpha\to\beta}, \tag{198}$$

where p_α and p_β denote the total incoming and outgoing 4-momenta. The factor $i(2\pi)^4\delta^{(4)}(p_\alpha - p_\beta)$ is purely kinematic; the quantity

$$M_{\alpha\to\beta}$$

is the **scattering amplitude**. This is the object measured in scattering experiments, and our theoretical goal is to compute it.

In previous lectures, we discussed general properties of the S-matrix such as unitarity and symmetry constraints. Here we focus on computing amplitudes.

2. LSZ Reduction and Correlation Functions

A key result is the **LSZ reduction theorem**, which states that the scattering amplitude is encoded in time-ordered correlation functions:

$$M_{\alpha\to\beta} \propto \langle \Omega | T(\phi(x_1) \cdots \phi(x_n)) \, | \Omega \rangle, \tag{199}$$

where $|\Omega\rangle$ is the true interacting vacuum and

$$n = (\text{number of incoming particles}) + (\text{number of outgoing particles}).$$

For example, in a $2 \to 2$ scattering process, $n = 4$. In a $2 \to 3$ process, $n = 5$.

We will later explain precisely why correlation functions determine scattering amplitudes. For now, our goal is:

$$\textbf{Compute } (199) \quad \Rightarrow \quad \textbf{obtain } M_{\alpha\to\beta} \quad \Rightarrow \quad \textbf{compare with experiment}.$$

3. Perturbation Theory in λ

The interacting theory cannot be solved exactly. Our only systematic method is to treat λ as small and expand all physical quantities in a power series. This is **perturbation theory**.

a) Perturbing the field operator. The equation of motion from (194) is

$$(\partial^2 - m^2)\phi = \frac{\lambda}{6}\phi^3. \tag{200}$$

We expand the field in powers of λ:

$$\phi = \phi_0 + \lambda\phi_1 + \lambda^2\phi_2 + \cdots . \tag{201}$$

Plugging this into (200) and matching powers of λ gives

$$(\partial^2 - m^2)\phi_0 = 0, \tag{202}$$
$$(\partial^2 - m^2)\phi_1 = \tfrac{1}{6}\phi_0^3, \tag{203}$$
$$\vdots \tag{204}$$

Thus ϕ_0 solves the free field equation, and ϕ_1, ϕ_2, ... can be solved in terms of ϕ_0.

b) Perturbing the vacuum. We also expand the interacting vacuum as

$$|\Omega\rangle = |0\rangle + \lambda|\psi_1\rangle + \lambda^2|\psi_2\rangle + \cdots , \tag{205}$$

where $|0\rangle$ is the free vacuum and the higher-order terms encode the effects of interactions.

Once ϕ and $|\Omega\rangle$ are expressed in terms of the free field ϕ_0 and $|0\rangle$, the correlation function (199) can be computed. Historically this "naive" method was used in the early days of QFT, but it becomes extremely complicated beyond the first few orders.

4. Two Advanced and Equivalent Approaches

4.1 The Interaction Picture

We split the Hamiltonian into free and interacting parts:

$$H = H_0 + H_I. \tag{206}$$

For (194), this corresponds to

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_I = \left(-\partial_\mu\phi\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2\right) + \left(-\tfrac{\lambda}{4!}\phi^4\right). \tag{207}$$

We then define interaction-picture states and operators:

$$|\Psi_I(t)\rangle = e^{iH_0t}\,|\Psi_S(t)\rangle, \tag{208}$$
$$O_I(t) = e^{iH_0t}\,O_S\,e^{-iH_0t}. \tag{209}$$

The operator evolution is controlled entirely by H_0, while the states evolve with H_I. This reshuffling allows a far more efficient formulation of perturbation theory (see Peskin §4.2).

4.2 The Path Integral Formulation

Path integrals provide an alternative formulation of quantum mechanics and QFT. It is conceptually clean and computationally powerful in field theory.

We illustrate the idea using nonrelativistic quantum mechanics with Hamiltonian

$$H = \frac{p^2}{2m} + V(x).$$

The starting point is the propagator:

$$K(x, t; x', t') \equiv \langle x, t \,|\, x', t'\rangle = \langle x|e^{-iH(t-t')}|x'\rangle. \tag{210}$$

This object fully determines time evolution:

$$\Psi(t, x) = \int dx'\, K(x, t; x', t')\,\Psi(t', x'). \tag{211}$$

Discretizing time. Divide $t' - t$ into N small intervals:

$$\Delta t = \frac{t - t'}{N}, \qquad N \to \infty.$$

Then

$$e^{-iH(t-t')} = \left(e^{-iH\Delta t}\right)^N.$$

Insert complete sets of position eigenstates:

$$K = \int dx_1 \cdots dx_{N-1} \prod_{i=0}^{N-1} \langle x_{i+1} | e^{-iH\Delta t} | x_i \rangle. \tag{212}$$

Each factor is approximated for small Δt using the BCH formula:

$$e^{-iH\Delta t} = e^{-i\frac{p^2}{2m}\Delta t} e^{-iV(x)\Delta t} \left(1 + O(\Delta t^2)\right).$$

A standard QM computation yields

$$\langle x_{i+1} | e^{-i\frac{p^2}{2m}\Delta t}\, e^{-iV(x)\Delta t} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp\left[\frac{im}{2}\left(\frac{x_{i+1} - x_i}{\Delta t}\right)^2 \Delta t - i\Delta t\, V(x_i)\right]. \tag{213}$$

Define $x_i = x(t_i)$ and identify the discrete action:

$$L = \frac{1}{2} m \dot{x}^2 - V(x).$$

Then the propagator becomes

$$K = \lim_{N\to\infty} \left(\frac{m}{2\pi i \Delta t}\right)^{N/2} \int dx_1 \cdots dx_{N-1} \exp\left[i\Delta t \sum_{i=0}^{N-1} L(x, \dot{x})\big|_{t_i}\right]. \tag{214}$$

Passing to the continuum limit gives the path integral:

$$K(x, t; x', t') = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(t)\, \exp\left[i \int_{t'}^{t} dt\, L(x, \dot{x})\right]. \tag{215}$$

Physical interpretation. The integral symbol $\mathcal{D}x(t)$ denotes integration over all possible paths $x(t)$ satisfying the fixed boundary conditions

$$x(t') = x', \quad x(t) = x.$$

Each path contributes with weight $\exp(iS)$, where

$$S = \int dt\, L(x, \dot{x})$$

is the classical action along that path. Classically, there is a *single* path connecting (x', t') and (x, t): the one satisfying the classical equations of motion. Quantum mechanically, *all* paths contribute. This is the essence of quantum fluctuations.

In the next lecture we will apply the path integral method directly to quantum field theory, and show how it provides an efficient framework for computing scattering amplitudes.

Lecture 13: Path Integrals in Nonrelativistic Quantum Mechanics

Path integrals provide an alternative formulation of quantum mechanics, fully equivalent to the Schrödinger and Heisenberg pictures. In nonrelativistic quantum mechanics, they mainly offer a more transparent conceptual picture (rather than a simpler computational tool), but in quantum field theory they become extremely powerful.

In this section we illustrate the construction of the path integral for a single particle moving in one dimension, with Hamiltonian

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x}). \tag{216}$$

1. Propagator as Starting Point

Instead of starting from the Schrödinger equation, the path integral formalism begins with the **propagator**

$$K(x,t;x',t') \equiv \langle x,t\,|\,x',t'\rangle, \tag{217}$$

defined in the Heisenberg picture as the transition amplitude for a particle initially in position eigenstate $|x',t'\rangle$ to be found in $|x,t\rangle$ at time t. Using time evolution,

$$K(x,t;x',t') = \langle x|e^{-i\hat{H}(t-t')}|x'\rangle. \tag{218}$$

If the wave function at time t' is $\Psi(t',x')$, then the wave function at time t is

$$\Psi(t,x) = \int dx'\, K(x,t;x',t')\, \Psi(t',x'). \tag{219}$$

Thus, knowing K is equivalent to solving the Schrödinger equation.

Our goal is to rewrite $K(x,t;x',t')$ as a sum over paths.

2. Time Slicing and Insertion of Completeness

Consider the time interval from t' to t. We divide it into N small segments:

$$t' = t_0 < t_1 < \cdots < t_{N-1} < t_N = t, \qquad \Delta t = \frac{t-t'}{N} \to 0. \tag{220}$$

Then

$$e^{-i\hat{H}(t-t')} = \left(e^{-i\hat{H}\Delta t}\right)^N. \tag{221}$$

Insert $N-1$ complete sets of position eigenstates:

$$\mathbf{1} = \int dx_i\, |x_i\rangle\langle x_i| \qquad (i = 1,\ldots,N-1), \tag{222}$$

to get

$$K(x,t;x',t') = \int dx_1 \cdots dx_{N-1} \prod_{i=0}^{N-1} \langle x_{i+1}|e^{-i\hat{H}\Delta t}|x_i\rangle, \tag{223}$$

where $x_N \equiv x$ and $x_0 \equiv x'$.

3. Short-Time Kernel and the Lagrangian

The short-time operator is

$$e^{-i\hat{H}\Delta t} = \exp\left[-i\Delta t\left(\frac{\hat{p}^2}{2m} + V(\hat{x})\right)\right]. \tag{224}$$

For small Δt, we use the Baker–Campbell–Hausdorff (BCH) formula to separate the exponent to leading order:

$$e^{-i\hat{H}\Delta t} \simeq e^{-i\Delta t\frac{\hat{p}^2}{2m}} e^{-i\Delta t V(\hat{x})}\left[1 + \mathcal{O}(\Delta t^2)\right]. \tag{225}$$

The $\mathcal{O}(\Delta t^2)$ corrections arise from commutators like $[\hat{p}^2,\hat{x}]$ in the BCH expansion

$$e^{A+B} = e^A e^B \exp\left(-\tfrac{1}{2}[A,B] + \cdots\right). \tag{226}$$

Thus the short-time matrix element is

$$\langle x_{i+1}|e^{-i\hat{H}\Delta t}|x_i\rangle \simeq \langle x_{i+1}|e^{-i\Delta t\frac{\hat{p}^2}{2m}} e^{-i\Delta t V(\hat{x})}|x_i\rangle. \tag{227}$$

A standard exercise in quantum mechanics (which we set as homework) gives the result

$$\langle x_{i+1}|e^{-i\Delta t\frac{\hat{p}^2}{2m}}e^{-i\Delta t V(\hat{x})}|x_i\rangle = \sqrt{\frac{m}{2\pi i\Delta t}}\exp\left[\frac{im}{2}\left(\frac{x_{i+1}-x_i}{\Delta t}\right)^2\Delta t - i\Delta t\, V(x_i)\right]. \tag{228}$$

Define

$$x(t_i) = x_i, \qquad \dot{x}(t_i) \approx \frac{x_{i+1}-x_i}{\Delta t}.$$

Then the exponent in (228) can be written as

$$i\Delta t\, L\big(x(t_i), \dot{x}(t_i)\big), \qquad L = \frac{1}{2}m\dot{x}^2 - V(x),$$

the classical Lagrangian.

4. Discrete Path Integral and Continuum Limit

Putting everything together,

$$K(x,t;x',t') = \lim_{N\to\infty}\left(\frac{m}{2\pi i\Delta t}\right)^{N/2}\int dx_1\cdots dx_{N-1}\exp\left[i\Delta t\sum_{i=0}^{N-1}L\big(x(t_i),\dot{x}(t_i)\big)\right]. \tag{229}$$

The sum approximates the time integral,

$$\Delta t\sum_{i=0}^{N-1}L(x,\dot{x}) \;\to\; \int_{t'}^{t}d\tilde{t}\, L(x,\dot{x}),$$

so in the continuum limit we write

$$K(x,t;x',t') = \int_{x(t')=x'}^{x(t)=x}\mathcal{D}x(t)\,\exp\left[i\int_{t'}^{t}d\tilde{t}\, L(x,\dot{x})\right]. \tag{230}$$

Here the symbolic measure $\mathcal{D}x(t)$ stands for

$$\mathcal{D}x(t) \sim \lim_{N\to\infty}\left(\frac{m}{2\pi i\Delta t}\right)^{N/2}dx_1\cdots dx_{N-1},$$

and the boundary conditions

$$x(t') = x', \qquad x(t) = x$$

are imposed on all paths.

Physical picture. The path integral has a simple interpretation: we integrate over *all* possible paths $x(t)$ connecting (x',t') and (x,t), and each path is weighted by the phase factor $\exp(iS)$, where the action is

$$S[x] = \int_{t'}^{t}d\tilde{t}\, L(x,\dot{x}).$$

In classical mechanics, a single trajectory is picked out by the equations of motion. In quantum mechanics, all trajectories contribute, with phases interfering constructively or destructively. This is a vivid way to see quantum fluctuations.

5. Hamiltonian Form of the Path Integral

An equivalent form uses both position and momentum:

$$K(x,t;x',t') = \int_{x(t')=x}^{x(t)=x}\mathcal{D}x(t)\,\mathcal{D}p(t)\,\exp\left[i\int_{t'}^{t}d\tilde{t}\,\big(p\,\dot{x} - H(p,x)\big)\right]. \tag{231}$$

For any given $x(t)$ and $p(t)$, we evaluate the phase $\int(p\dot{x}-H)\,dt$ and then sum over all such phase-space paths. One can show that this Hamiltonian form is equivalent to the Lagrangian form derived above.

6. Example: Free Particle Amplitude

Consider a free particle with action

$$S[q] = \int dt\, \frac{1}{2} m\dot{q}^2. \tag{232}$$

We look at the amplitude

$$Z_T = \langle 0, T | 0, 0 \rangle,$$

the amplitude for a particle starting at $q = 0$ at $t = 0$ to return to $q = 0$ at time $t = T$. Using the discrete form (229), one can show

$$Z_T = \left(\frac{m}{2\pi i T} \right)^{1/2}, \tag{233}$$

which agrees with the result obtained by solving the Schrödinger equation in the usual operator formalism.

7. Gaussian Functional Integral Viewpoint

There is also a more abstract, but very useful, way of looking at the free particle path integral. Start from

$$S[q] = \frac{1}{2} m \int_0^T dt\, \dot{q}^2 = \frac{1}{2} \int_0^T dt\, q(t) \left(-m\partial_t^2 \right) q(t), \tag{234}$$

where we integrated by parts and used $q(0) = q(T) = 0$ so that boundary terms vanish. Introducing a second time variable t',

$$S[q] = \frac{1}{2} \int_0^T dt\, dt'\, q(t)\, K(t, t')\, q(t'), \tag{235}$$

with

$$K(t, t') = -m\, \delta(t - t')\, \partial_{t'}^2. \tag{236}$$

We treat t and t' as indices, so $q(t)$ is like a vector and K like a matrix. Then the path integral

$$Z_T = \int \mathcal{D}q(t)\, \exp\left[\frac{i}{2} \int dt\, dt'\, q(t)\, K(t, t')\, q(t') \right] \tag{237}$$

is the infinite-dimensional analog of a Gaussian integral

$$\int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}x^2} = \sqrt{\frac{2\pi}{a}}, \tag{238}$$

$$\int d^n X\, e^{-\frac{1}{2} X_m A_{mn} X_n} = \frac{(2\pi)^{n/2}}{\sqrt{\det A}}. \tag{239}$$

By analogy, we expect

$$Z_T = \frac{C}{\sqrt{\det K}}, \tag{240}$$

where C is some (divergent) normalization constant and $\det K$ is the functional determinant (the product of eigenvalues) of the operator K.

The eigenvalue problem

$$\int_0^T dt'\, K(t, t')\, f_j(t') = \lambda_j f_j(t) \tag{241}$$

with boundary conditions $f_j(0) = f_j(T) = 0$ leads to eigenfunctions and eigenvalues

$$f_j(t) = \sin\left(\frac{\pi j t}{T} \right), \qquad \lambda_j = m\frac{j^2\pi^2}{T^2}, \qquad j = 1, 2, \ldots$$

so formally

$$\det K = \prod_{j=1}^{\infty} m\frac{j^2\pi^2}{T^2}.$$

Although both C and $\det K$ are divergent, physical quantities are typically ratios of such amplitudes in which these overall constants cancel. In practice, we almost never need to evaluate these determinants explicitly; the main value of this viewpoint is conceptual and as a bridge to quantum field theory.

8. Remarks

- The path integral is a new formulation of quantum mechanics, but it is fully *equivalent* to the Schrödinger formulation. One can show that the propagator $K(x, t; x', t')$ defined by the path integral satisfies the Schrödinger equation

$$i\frac{\partial}{\partial t}K(x, t; x', t') = \left[-\frac{1}{2m}\partial_x^2 + V(x)\right] K(x, t; x', t').$$

- Conceptually, the contrast with classical mechanics is sharp: classically, a single path extremizes the action; quantum mechanically, we sum over *all* paths with phase weights e^{iS}. This picture is the foundation for the path integral formulation of quantum field theory.

Lecture 14: Path Integral Formalism for QFT and Time-Ordered Correlation Functions

Last time, we introduced the path integral formulation of non-relativistic quantum mechanics. Classically, specifying the initial point (t', x') and final point (t, x) determines a unique trajectory. Quantum mechanically, the transition amplitude $\langle x, t|x', t'\rangle$ is obtained by summing over *all* possible paths between (t', x') and (t, x):

$$\langle x, t|x', t'\rangle = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(t)\, e^{\frac{i}{\hbar}S[x(t)]}. \tag{242}$$

Here, $S[x(t)]$ is a functional of the full trajectory:

$$S[x(t)] = \int_{t'}^{t} d\tilde{t}\left(\frac{1}{2}m\dot{x}^2 - V(x)\right), \tag{243}$$

with fixed endpoints $x(t') = x'$ and $x(t) = x$.

The shorthand notation $\mathcal{D}x(t)$ is defined as the continuum limit of a discrete product of ordinary integrals:

$$\int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(t) \equiv \lim_{N\to\infty}\left(\frac{m}{2\pi i\Delta t}\right)^{N/2}\int dx_1\cdots dx_{N-1}. \tag{244}$$

We divide the interval $t' \to t$ into N pieces; at each intermediate time t_i we integrate over all possible positions x_i. This is the fundamental definition of the path integral in quantum mechanics.

Example: The Free Particle

Consider

$$S = \int dt\, \frac{1}{2}m\dot{x}^2. \tag{245}$$

In this case the path integral is Gaussian. Schematically, we have

$$\int \mathcal{D}x(t)\exp\left[\frac{i}{2}\int dtdt'\, x(t)K(t,t')x(t')\right], \tag{246}$$

where $K(t, t')$ is a differential operator. The result of a Gaussian integral is

$$Z = \frac{C}{\sqrt{\det K}}. \tag{247}$$

One may obtain the same result by taking the continuum limit of (244).

This is the path integral for one degree of freedom. For finitely many degrees of freedom, we simply integrate over all coordinates. The generalization to fields which have infinitely many degrees of freedom introduces no conceptual difficulty: the field value at each spatial point becomes a dynamical variable.

1. Path Integral for Quantum Field Theory

In quantum mechanics, the dynamical variable is $\hat{x}(t)$. In quantum field theory, the dynamical variable becomes the field operator $\hat{\phi}(t, \mathbf{x})$. The analogue of the transition amplitude is

$$\langle \phi(\mathbf{x}), t \,|\, \phi'(\mathbf{x}), t'\rangle = \langle \phi|e^{-iH(t-t')}|\phi'\rangle. \tag{248}$$

We slice the time interval, insert complete sets of field eigenstates $|\phi(\mathbf{x})\rangle$, and follow exactly the same steps as in quantum mechanics. The result is

$$\boxed{\langle \phi, t|\phi', t'\rangle = \int_{\phi(t')=\phi'}^{\phi(t)=\phi} \mathcal{D}\phi(t,\mathbf{x})\, e^{\frac{i}{\hbar}S[\phi]}} \tag{249}$$

where

$$S[\phi] = \int d^4x\left(-\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi)\right). \tag{250}$$

The functional measure is interpreted as

$$\mathcal{D}\phi(t,\mathbf{x}) = \prod_{t,\mathbf{x}} d\phi(t,\mathbf{x}). \tag{251}$$

Every spatial point labels one degree of freedom.

Key conceptual point. Aside from the extra label $\mathbf{x}$, the field-theory path integral is *identical* to the quantum mechanical one. A free scalar field yields a Gaussian integral whose kernel K is simply a higher-dimensional differential operator.

2. Time-Ordered Correlation Functions in QM

Our goal in QFT is to compute

$$\langle\Omega|T\big(\phi(x_1)\cdots\phi(x_n)\big)|\Omega\rangle, \tag{252}$$

the vacuum expectation values of time-ordered products. To understand how path integrals compute such objects, it is instructive to first review the corresponding story in QM. The QM analogue of (252) is

$$G_n = \langle 0|T\{\hat{x}(t_1)\cdots\hat{x}(t_n)\}|0\rangle. \tag{253}$$

2.1 How operators appear inside a path integral

Consider first a single operator insertion:

$$\langle x,t|\hat{x}(t_1)|x',t'\rangle = \int dx_1\,\langle x,t|x_1,t_1\rangle\,x_1\,\langle x_1,t_1|x',t'\rangle. \tag{254}$$

Using the path integral expressions for each factor, one obtains

$$\langle x,t|\hat{x}(t_1)|x',t'\rangle = \int_{(x',t')}^{(x,t)} \mathcal{D}x(t)\,x(t_1)\,e^{\frac{i}{\hbar}S[x(t)]}. \tag{255}$$

Similarly, two insertions give

$$\langle x,t|\hat{x}(t_1)\hat{x}(t_2)|x',t'\rangle = \int \mathcal{D}x(t)\,x(t_1)x(t_2)\,e^{\frac{i}{\hbar}S[x(t)]}, \tag{256}$$

independent of whether $t_1 > t_2$ or not. The path integral naturally orders time variables because paths evolve forward in time. Thus,

$$\boxed{\langle x,t|T\{\hat{x}(t_1)\cdots\hat{x}(t_n)\}|x',t'\rangle = \int \mathcal{D}x(t)\,x(t_1)\cdots x(t_n)e^{\frac{i}{\hbar}S[x(t)]}}. \tag{257}$$

2.2 Selecting the Vacuum Using the $i\epsilon$ Trick

To compute the vacuum correlators

$$G_n = \langle 0|T(x_1\cdots x_n)|0\rangle,$$

we insert resolutions of identity at $t\to+\infty$ and $t'\to-\infty$:

$$1 = \int dx\,|x,t\rangle\langle x,t|.$$

Expanding in energy eigenstates and introducing a small damping factor $E_n \to E_n(1-i\epsilon)$, we find

$$\lim_{t\to+\infty}\langle x,t|n\rangle = \begin{cases} e^{-iE_0(1-i\epsilon)t}\psi_0(x), & n=0,\\ 0, & n\neq 0.\end{cases}$$

Thus only the ground state survives:

$$\lim_{t\to+\infty,t'\to-\infty}\langle x,t|X|x',t'\rangle = \psi_0(x)\psi_0^*(x')e^{-iE_0(t-t')}\langle 0|X|0\rangle. \tag{258}$$

Taking the ratio with the propagator without operator insertions eliminates wave functions and the ground-state phase:

$$\boxed{G_n = \frac{\displaystyle\lim_{t\to\infty,t'\to-\infty}\langle x,t|X|x',t'\rangle}{\displaystyle\lim_{t\to\infty,t'\to-\infty}\langle x,t|x',t'\rangle}}. \tag{259}$$

This ratio removes the overall normalization constant C from the Gaussian path integral.

Lecture 15: Generating Functionals and Correlation Functions

Review: Correlation Functions and Path Integrals

Last time, we discussed how to compute vacuum correlation functions

$$G_n(t_1, \ldots, t_n) = \langle 0| T(\hat{x}(t_1) \cdots \hat{x}(t_n)) |0\rangle \tag{260}$$

in a single-particle quantum mechanical system described by

$$\mathcal{L} = \frac{1}{2} m\dot{x}^2 - V(x). \tag{261}$$

Using the path integral formulation, we derived the general expression

$$G_n = \frac{\int \mathcal{D}x(t)\, X\, e^{iS_\epsilon[x(t)]}}{\int \mathcal{D}x(t)\, e^{iS_\epsilon[x(t)]}}, \tag{262}$$

where $X = x(t_1) \cdots x(t_n)$, and the path integral is defined with boundary conditions

$$x(-\infty) = x(+\infty) = 0.$$

To make the vacuum well-defined, we introduced the standard $i\epsilon$ prescription,

$$H \to H(1 - i\epsilon), \qquad 0 < \epsilon \ll 1, \tag{263}$$

which induces a small imaginary part in the action. After all calculations are completed, we take the limit $\epsilon \to 0$.

Motivation: Why Generating Functionals?

Although Eq. (262) allows us to compute any n-point function, doing this separately for each n is inefficient. Instead, we introduce a powerful tool: the **generating functional**. To motivate this idea, let us first consider a simple one-dimensional example.

Warm-Up: Generating Functions in Ordinary Integrals

Suppose we want to compute

$$Z_n = \int dx\, e^{i\lambda f(x)}\, x^n \tag{264}$$

for many values of n. Rather than computing each integral separately, we define

$$Z(a) = \int dx\, e^{i\lambda f(x) + iax}. \tag{265}$$

Taking derivatives with respect to a brings down powers of x:

$$Z_n = \frac{1}{i^n} \left. \frac{\partial^n Z(a)}{\partial a^n} \right|_{a=0}. \tag{266}$$

Equivalently, $Z(a)$ admits the expansion

$$Z(a) = \sum_{n=0}^{\infty} \frac{i^n}{n!} Z_n a^n, \tag{267}$$

so $Z(a)$ is the **generating function** of the moments Z_n.

Generating Functionals in Quantum Mechanics

We now generalize this idea from ordinary integrals to path integrals. The generating functional is defined as

$$Z[J] = \int \mathcal{D}x(t) \exp\left(iS[x(t)] + i \int dt\, J(t) x(t) \right), \tag{268}$$

where $J(t)$ is an external source. The functional derivative obeys

$$\frac{\delta J(t')}{\delta J(t)} = \delta(t - t'). \tag{269}$$

Taking a functional derivative of $Z[J]$ gives

$$\frac{\delta Z[J]}{\delta J(t)} = i \int \mathcal{D}x(t)\, x(t)\, e^{iS[x]+i\int dt J(t)x(t)}. \tag{270}$$

Defining

$$Z_0 \equiv Z[J=0] = \int \mathcal{D}x(t)\, e^{iS[x(t)]}, \tag{271}$$

we obtain

$$\langle 0|\hat{x}(t)|0\rangle = \frac{1}{iZ_0} \frac{\delta Z[J]}{\delta J(t)}\bigg|_{J=0}. \tag{272}$$

More generally,

$$G_n(t_1, \ldots, t_n) = \frac{1}{i^n Z_0} \frac{\delta^n Z[J]}{\delta J(t_1) \cdots \delta J(t_n)}\bigg|_{J=0}. \tag{273}$$

Thus, once $Z[J]$ is known, *all* correlation functions follow by differentiation.

Time-Ordered Interpretation

The generating functional satisfies

$$\frac{Z[J]}{Z_0} = \langle 0|T \exp\left(i \int dt\, J(t)\hat{x}(t)\right)|0\rangle. \tag{274}$$

Expanding the exponential,

$$\frac{Z[J]}{Z_0} = \sum_{n=0}^{\infty} \frac{i^n}{n!} \int dt_1 \cdots dt_n\, G_n(t_1, \ldots, t_n)\, J(t_1) \cdots J(t_n). \tag{275}$$

Example: Harmonic Oscillator

Consider the harmonic oscillator

$$S = \int_{-\infty}^{\infty} \left(\frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega_0^2 x^2\right) dt. \tag{276}$$

We want the vacuum-to-vacuum path integral, so we slightly damp excited states by the standard replacement

$$H \;\to\; (1-i\epsilon)\, H, \qquad 0 < \epsilon \ll 1,$$

because then

$$e^{-iHT} \;\to\; e^{-i(1-i\epsilon)HT} = e^{-iHT}\, e^{-\epsilon HT},$$

and the factor $e^{-\epsilon HT}$ suppresses higher-energy states as $T \to \infty$. Start from the harmonic oscillator Hamiltonian

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega_0^2 x^2, \qquad H_\epsilon = (1-i\epsilon)H.$$

The Lagrangian is obtained by the Legendre transform:

$$L_\epsilon = p\dot{x} - H_\epsilon, \qquad \dot{x} = \frac{\partial H_\epsilon}{\partial p} = (1-i\epsilon)\frac{p}{m}.$$

So

$$p = \frac{m}{1-i\epsilon}\dot{x} \;=\; m(1+i\epsilon)\dot{x} \quad \text{(to first order in } \epsilon\text{)}.$$

Plug this into L_ϵ:

$$\begin{aligned} L_\epsilon &= p\dot{x} - (1-i\epsilon)\left(\frac{p^2}{2m} + \frac{1}{2}m\omega_0^2 x^2\right) \\ &= \frac{m}{2(1-i\epsilon)}\dot{x}^2 - \frac{1}{2}(1-i\epsilon)m\omega_0^2 x^2 \\ &\simeq \frac{1}{2}m(1+i\epsilon)\dot{x}^2 - \frac{1}{2}m\omega_0^2(1-i\epsilon)x^2, \end{aligned}$$

where we used $\frac{1}{1-i\epsilon} \simeq 1 + i\epsilon$. Now rewrite the kinetic term by integration by parts (boundary terms vanish since $x(\pm\infty) = 0$):

$$\int dt\, \frac{1}{2} m(1+i\epsilon)\dot{x}^2 = -\int dt\, \frac{1}{2} m(1+i\epsilon)\, x\, \partial_t^2 x.$$

Therefore the quadratic operator in the action is

$$S_\epsilon = \int dt\, L_\epsilon = -\frac{1}{2} \int dt\, m\, x \Big[(1+i\epsilon)\partial_t^2 + \omega_0^2(1-i\epsilon) \Big] x.$$

Since ϵ is infinitesimal, we keep only the fact that it produces a *negative imaginary part* in the inverse propagator (the precise coefficient is irrelevant, only the sign matters). Thus we write the standard form

$$S_\epsilon = -\frac{1}{2} \int dt\, m\, x \left(\partial_t^2 + \omega_0^2 - i\epsilon \right) x,$$

or (often setting $m = 1$ in the notes)

$$L_\epsilon = -\frac{1}{2}\, x \left(\partial_t^2 + \omega_0^2 - i\epsilon \right) x. \tag{277}$$

This $-i\epsilon$ is exactly what shifts the poles to define the Feynman propagator. The action can be written in quadratic form:

$$S[x] = -\frac{1}{2} \int dt\, dt'\, x(t) K(t,t') x(t'), \tag{278}$$

with

$$K(t,t') = \delta(t-t') \left(\partial_t^2 + \omega_0^2 - i\epsilon \right). \tag{279}$$

Gaussian Functional Integral

The vacuum functional is

$$Z_0 = \int \mathcal{D}x\, \exp\!\left(-\frac{i}{2} x \cdot K \cdot x \right) = \frac{C}{\sqrt{\det K}}. \tag{280}$$

Including the source,

$$Z[J] = \frac{C}{\sqrt{\det K}} \exp\!\left(\frac{i}{2} J \cdot K^{-1} \cdot J \right), \tag{281}$$

where

$$\int dt'\, K(t,t') K^{-1}(t',t'') = \delta(t-t''). \tag{282}$$

Thus,

$$\frac{Z[J]}{Z_0} = \exp\!\left(\frac{i}{2} \int dt\, dt'\, J(t) K^{-1}(t,t') J(t') \right). \tag{283}$$

Feynman Propagator and Wick's Theorem

The two-point function is

$$G_F(t,t') = \langle 0 | T(\hat{x}(t)\hat{x}(t')) | 0 \rangle = -iK^{-1}(t,t'). \tag{284}$$

K^{-1} satisfies

$$\left(\partial_t^2 + \omega_0^2 - i\epsilon \right) K^{-1}(t,t') = \delta(t-t'). \tag{285}$$

In momentum space,

$$G_F(\omega) = \frac{i}{\omega^2 - \omega_0^2 + i\epsilon}. \tag{286}$$

All odd n-point functions vanish:

$$G_n = 0 \qquad (n \text{ odd}), \tag{287}$$

while even-point functions are given by sums over pairings:

$$G_n = \sum_{\text{all pairings}} \prod G_F(t_i, t_j). \tag{288}$$

This is precisely **Wick's theorem**. In the path integral approach, it follows immediately from the Gaussian nature of $Z[J]$.

Outlook

With this machinery developed in quantum mechanics, we are now fully prepared to generalize the generating functional formalism to quantum field theory.

Lecture 16: Time-Ordered Correlation Functions and Feynman Diagrams

To generalize from quantum mechanics to quantum field theory, the procedure is almost entirely notational. The conceptual structure is identical: we simply replace the dynamical variable $x(t)$ by a field $\phi(x)$ defined over spacetime.

The n-point function in field theory is defined as

$$G_n(x_1, \ldots, x_n) = \langle\Omega|T(\phi(x_1)\cdots\phi(x_n))\,|\Omega\rangle \equiv \langle\Omega|X|\Omega\rangle, \tag{289}$$

where $x_i = (t_i, \vec{x}_i)$ are spacetime points and $|\Omega\rangle$ denotes the interacting vacuum. Using the path integral, this can be written as

$$G_n(x_1, \ldots, x_n) = \frac{\int \mathcal{D}\phi\; X\; e^{iS[\phi]}}{\int \mathcal{D}\phi\; e^{iS[\phi]}}. \tag{290}$$

Boundary conditions. We impose

$$\phi(t, \vec{x}) \to 0 \quad \text{as} \quad t \to \pm\infty, \qquad \phi(t, \vec{x}) \to 0 \quad \text{as} \quad |\vec{x}| \to \infty.$$

Physically, this means that the vacuum is unexcited at asymptotic times and far away in space. This ensures the path integral is well defined.

Generating Functional in Field Theory

As in quantum mechanics, it is convenient to introduce a generating functional:

$$Z[J] = \int \mathcal{D}\phi\; \exp\!\left(iS[\phi] + i\int d^4x\, J(x)\phi(x)\right). \tag{291}$$

Defining $Z_0 = Z[J = 0]$, we have

$$\frac{Z[J]}{Z_0} = \langle\Omega|T\exp\!\left(i\int d^4x\, J(x)\phi(x)\right)|\Omega\rangle. \tag{292}$$

Expanding the exponential shows that $Z[J]$ generates all time-ordered correlation functions. Functional differentiation gives

$$G_n(x_1, \ldots, x_n) = \frac{1}{i^n}\frac{\delta^n}{\delta J(x_1)\cdots\delta J(x_n)}\frac{Z[J]}{Z_0}\bigg|_{J=0}.$$

This formula applies to *any* scalar field theory, including interacting ones.

Free Scalar Field Theory

Consider the free scalar field

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2. \tag{293}$$

The action can be written in quadratic form:

$$S_0 = -\frac{1}{2}\int d^4x\, d^4x'\; \phi(x)\,K(x, x')\,\phi(x') \equiv -\frac{1}{2}\phi\cdot K\cdot\phi, \tag{294}$$

with

$$K(x, x') = \left(-\partial^2 + m^2 - i\epsilon\right)\delta^{(4)}(x - x'). \tag{295}$$

The generating functional becomes a Gaussian integral:

$$Z[J] = \int \mathcal{D}\phi\; e^{-\frac{i}{2}\phi\cdot K\cdot\phi + iJ\cdot\phi} = \frac{C}{\sqrt{\det K}}\exp\!\left(\frac{i}{2}J\cdot K^{-1}\cdot J\right), \tag{296}$$

and

$$Z_0 = \frac{C}{\sqrt{\det K}}. \tag{297}$$

Therefore,

$$\frac{Z[J]}{Z_0} = \exp\!\left(-\frac{1}{2}\int d^4x\, d^4x'\; J(x)\,G_F(x, x')\,J(x')\right), \tag{298}$$

where

$$K^{-1}(x, x') = iG_F(x, x'). \tag{299}$$

All correlation functions follow from Wick's theorem. For example, the four-point function is

$$G_4 = G_F(x_1, x_2)G_F(x_3, x_4) + G_F(x_1, x_3)G_F(x_2, x_4) + G_F(x_1, x_4)G_F(x_2, x_3). \tag{300}$$

Interacting Theory and Perturbation Expansion

We now move beyond free field theory and consider an interacting scalar field. As a concrete example, we take

$$\mathcal{L} = \mathcal{L}_0 - \frac{\lambda}{4!}\phi^4 \equiv \mathcal{L}_0 + \mathcal{L}_I, \tag{301}$$

where $\mathcal{L}_0$ is the free Lagrangian and $\mathcal{L}_I$ describes the interaction. Correspondingly, the action splits into two parts,

$$S[\phi] = S_0[\phi] + S_I[\phi], \qquad S_I = \int d^4x\, \mathcal{L}_I. \tag{302}$$

Since $\mathcal{L}_I$ contains no time derivatives, it is related to the interaction Hamiltonian by

$$S_I = -\int dt\, H_I. \tag{303}$$

Exact expression for correlation functions

The exact time-ordered n-point function in the interacting theory is

$$\langle\Omega|TX|\Omega\rangle = \frac{\int \mathcal{D}\phi\; X\, e^{iS_0[\phi]} e^{iS_I[\phi]}}{\int \mathcal{D}\phi\; e^{iS_0[\phi]} e^{iS_I[\phi]}}, \tag{304}$$

where $|\Omega\rangle$ is the interacting vacuum and $X = \phi(x_1)\cdots\phi(x_n)$. This expression is exact, but in practice it cannot be evaluated in closed form because the interaction term makes the path integral non-Gaussian.

Reduction to free-theory expectation values

The key idea of perturbation theory is to treat S_I as small and expand it. Before expanding, we rewrite (304) as

$$\langle\Omega|TX|\Omega\rangle = \frac{\langle 0|T\big(Xe^{iS_I}\big)\,|0\rangle}{\langle 0|T(e^{iS_I})\,|0\rangle}, \tag{305}$$

where $|0\rangle$ is the vacuum of the free theory.

This formula coincides with Sec4.3.1 of Peskin and Schroeder. It expresses interacting-theory correlation functions entirely in terms of free-theory expectation values.

Perturbative expansion

We now expand the exponential in powers of λ. To first order,

$$G_n = \frac{\langle 0|TX|0\rangle - i\int dt\,\langle 0|T(XH_I)|0\rangle + \cdots}{1 - i\int dt\,\langle 0|TH_I|0\rangle + \cdots}. \tag{306}$$

Expanding the denominator and keeping terms to first order gives

$$G_n = G_n^{(0)} + iG_n^{(0)}\int dt\,\langle 0|TH_I|0\rangle - i\int dt\,\langle 0|T(XH_I)|0\rangle + \cdots, \tag{307}$$

where $G_n^{(0)} = \langle 0|TX|0\rangle$ is the free-theory n-point function.

Important point. All expectation values appearing here are taken in the free theory. Therefore, they can be evaluated systematically using Wicks theorem.

Feynman Diagrams

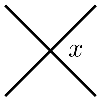
Although Wicks theorem provides a systematic method, the number of terms grows rapidly with perturbative order. Feynman diagrams provide a graphical way to organize and simplify this expansion.

Basic graphical elements

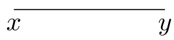
For the interaction

$$\mathcal{L}_I = -\frac{\lambda}{4!}\phi^4,$$

each factor of S_I produces a four-leg interaction vertex,



which corresponds to an integration $\int d^4x$ over the vertex position. The free propagator is represented by a line,



$$= G_F^{(0)}(x-y).$$

Example: two-point function at first order

Let us apply this to the two-point function

$$G_2(x_1 - x_2) = \langle\Omega|T\phi(x_1)\phi(x_2)|\Omega\rangle. \tag{308}$$

Expanding to first order in λ gives

$$\begin{aligned} G_2(x_1-x_2) = G_F^{(0)}(x_1-x_2) + i\frac{\lambda}{4!}G_F^{(0)}(x_1-x_2)\int d^4x\langle 0|T\phi^4(x)|0\rangle \\ -i\frac{\lambda}{4!}\int d^4x\langle 0|T\left(\phi(x_1)\phi(x_2)\phi^4(x)\right)|0\rangle + O(\lambda^2). \end{aligned} \tag{309}$$

The zeroth-order term is simply the free propagator. By translation invariance, it depends only on $x_1 - x_2$.

Wick contractions and cancellations

Applying Wicks theorem,

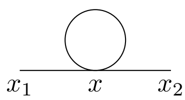
$$\begin{aligned} G_F(x_1-x_2) = G_F^{(0)}(x_1-x_2) + i\frac{\lambda}{4!}G_F^{(0)}(x_1-x_2)\,3\int d^4x\, G_F^{(0)}(0)G_F^{(0)}(0) \\ -i\frac{\lambda}{4!}\,4\times 3\int d^4x\, G_F^{(0)}(x_1-x)G_F^{(0)}(x_2-x)G_F^{(0)}(0) + O(\lambda^2). \end{aligned} \tag{310}$$

The first correction term cancels against the denominator in (305). As a result, the interacting two-point function becomes

$$G_F(x_1-x_2) = G_F^{(0)}(x_1-x_2) - i\frac{\lambda}{4!}\,4\times 3\int d^4x\, G_F^{(0)}(x_1-x)G_F^{(0)}(x_2-x)G_F^{(0)}(0) + O(\lambda^2). \tag{311}$$

This expression is represented diagrammatically as

$$-\frac{i\lambda}{2}$$



$$+\ O(\lambda^2).$$

Higher orders and combinatorics

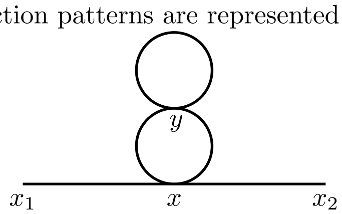
At second order, terms such as

$$\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)^2\int d^4x\, d^4y\ \langle 0|T\phi(x_1)\phi(x_2)\phi^4(x)\phi^4(y)|0\rangle \tag{312}$$

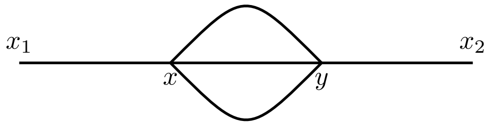
appear.

Instead of writing out all Wick contractions explicitly, we classify them by drawing diagrams. All possible contraction patterns are represented by the diagrams

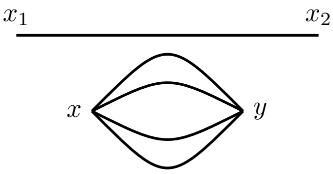
(a)



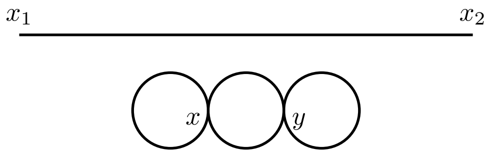
with the factor $\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)\cdot 2\cdot 4\cdot 3\cdot 4\cdot 3 = -\lambda^2\left(\frac{1}{2}\right)^2$.

(b) 

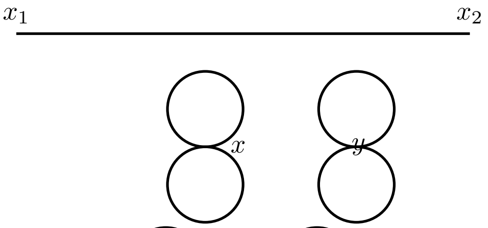
with the factor $\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)\cdot 2\cdot 4\cdot 4\cdot 3\cdot 2 = -\frac{\lambda^2}{3!}$

(c) 

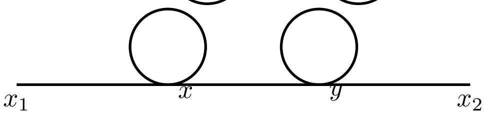
with the factor $\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)\cdot 4\cdot 3\cdot 2 = -\frac{\lambda^2}{2}\frac{1}{4!}$.

(d) 

with the factor $\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)\cdot 2\cdot 3\cdot 4\cdot 3 = -\frac{\lambda^2}{16}$.

(e) 

with the factor $\frac{1}{2!}\left(-\frac{i\lambda}{4!}\right)\cdot 3^2 = -\frac{\lambda^2}{2}\left(\frac{1}{8}\right)^2$

(f) 

Each diagram comes with a combinatorial (symmetry) factor, determined by counting equivalent contractions.

Key message for students.

- Interacting correlation functions are reduced to free ones by perturbation theory.
- Wicks theorem turns operator products into sums of propagators.
- Feynman diagrams encode contraction patterns and symmetry factors automatically.
- Drawing diagrams first is far more efficient than writing algebraic expressions.

In the next lecture, we will formalize this procedure and extract systematic Feynman rules.